The economic consequences of geopolitical fragmentation: Evidence from the Cold War*

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September 5, 2026

Abstract

This article constructs new data on trade flows involving socialist economies, missing from standard datasets, and shows that trade barriers between Eastern and Western Europe were much larger and more volatile than previously available information suggest. Using primary historical sources, we reconstruct bilateral and domestic trade flows for Eastern European countries, recovering a complete trade matrix for the Cold War period. These data are essential to identify trade barriers in a structural gravity framework. Our estimates suggest that trade barriers between Eastern and Western Europe were large and varied substantially over time, declining during periods of easing tensions and rising with renewed confrontation. The results provide a benchmark for how trade barriers vary with geopolitical tensions.

Keywords: International trade, Cold War, Iron Curtain, geopolitical fragmentation, trade costs of borders

JEL codes: F13, F14, N74

*We thank Rodrigo Peña and Maximilano Moreno-López for excellent research assistance and Marina Diakonova for her help in locating the digitized copies of the statistical reports on foreign trade of the USSR. We thank Lorenzo Caliendo, Douglas Irwin, Pinelopi Goldberg, Giovanni Maggi, David Atkin, Esteban Rossi-Hansberg, Ethan Ilzetzki, Kris Mitchener, Evi Pappa, Markus Lampe, Ben Shepherd, Joschka Wanner, Costanza Bosone and audiences at the 2026 Kuznets Mini-Conference at Yale, the XXV Conference on International Economics, Alicante, the Workshop on European Integration, Economic Convergence and Productivity at the College of Europe, Bruges, the PSE Macro Days 2024, the Banco de España research seminar, the Lille Économie Management research seminar, the Trade Policy Research Forum, and the XIV International Congress of the Spanish Association of Economic History for helpful comments and suggestions. Heid gratefully acknowledges financial support from the Spanish Ministry of Universities (Ministerio de Universidades, Plan de Recuperación, Transformación y Resiliencia) under María Zambrano contract MAZ/2021/04(UP2021-021) financed by the European Union - NextGenerationEU, from PID2020-114646RB-C42 funded by MCIN-AEI/10.13039/501100011033 (a project from the Ministerio de Ciencia e Innovación (MCIN), Agencia Estatal de Investigación (AEI), Spain), from the Spanish Ministry of Science and Innovation and the EU (PID2023-147397NB-I00, FEDER 2021-2027), from Prometeo project CIPROM/2022/50 financed by the Generalitat Valenciana, and from the Plan de Promoción de la Investigación de la Universitat Jaume I, UJI-B2022-36-(22I587). The funding sources had no involvement in the writing of this paper nor the decision to submit it for publication. Declarations of interest: none. The views expressed in this paper are those of the authors and do therefore not necessarily reflect those of the Banco de España or the Eurosystem. The date of the first version of this paper is 4 April 2024.

1 Introduction

How large are the trade barriers created by geopolitical fragmentation? This question has become increasingly important as geopolitical tensions reshape international economic relations and raise concerns about the fragmentation of global trade. While recent work documents changes in trade patterns associated with geopolitical alignment, there is limited evidence on the magnitude of the underlying trade barriers and how they evolve over time.

The Cold War provides a useful setting to study this question. Europe remained divided into an Eastern and a Western bloc for several decades, with stable geopolitical alignment but substantial variation in tensions over time. The countries separated by the Iron Curtain were geographically close and had strong economic links prior to the division, leaving substantial scope for trade in the absence of barriers.

Recovering these barriers, however, requires accurate measurement of trade flows during the Cold War. This is difficult because standard international datasets provide incomplete coverage of trade involving socialist economies, particularly within the Eastern bloc. One of the largest trade relationships across the Cold War divide—trade between East and West Germany—is largely absent from standard international datasets because the two German states classified these flows as *Innerdeutscher Handel* (intra-German trade) rather than international trade. More broadly, many economically important Eastern-bloc trade flows are either missing or observed only through mirror statistics reported by trading partners. As a result, existing datasets do not provide a complete and internally coherent bilateral trade matrix for the Cold War period, making it difficult to recover structurally interpretable measures of trade barriers between Eastern and Western Europe..

We address this problem by reconstructing the bilateral and domestic flows needed to estimate trade barriers between Eastern and Western Europe within a structural gravity framework. Combining primary historical sources for Eastern European economies with existing international data, we recover a more complete trade matrix for the Cold War period. This allows us to trace the evolution of these barriers over time.

We find that trade barriers between Eastern and Western Europe are large and strongly time-varying throughout the Cold War. They decline during periods of easing tensions, rise again during the renewed geopolitical escalation of the early 1980s, and fall sharply during the final phase of the Cold War. Standard international datasets substantially understate their level and and attenuate changes over time by omitting economically important flows involving socialist economies. As a result, they give a misleading picture of trade integration during the Cold War.

More broadly, the results show that geopolitical fragmentation can generate large and persistent changes in trade barriers even in the absence of direct military conflict or formal trade wars.

The Cold War therefore provides a benchmark for understanding how geopolitical tensions affect economic integration through changes in the costs of trading across borders. This perspective is particularly relevant in light of current debates about how geopolitical tensions and government policies affect international trade, as they change the costs of trading across countries and lead firms to adjust who they trade with and how much they trade. By recovering a time-varying measure of these barriers, the paper provides a framework to quantify how these changes in trade costs affect trade patterns and economic integration.

Contribution to the literature

This paper contributes to work on geopolitics and international trade by measuring the trade barriers associated with one of the central episodes of twentieth-century geopolitical fragmentation. A large literature shows that political alignment, conflict, and military rivalry affect trade flows (Gowa, 1994; Martin et al., 2008; Glick and Taylor, 2010; Acemoglu and Yared, 2010; Berger et al., 2013; Fuchs and Klann, 2013). Recent work has renewed this question in the context of rising geopolitical tensions and the reorientation of trade relationships across countries (Aiyar et al., 2023a,b; Campos et al., 2023; Alfaro et al., 2024; Gopinath et al., 2025). We contribute by estimating the time path of trade barriers between Eastern and Western Europe during the Cold War. This requires new data, since standard international sources provide incomplete coverage of trade involving Eastern European economies, especially within the Eastern bloc and across the East–West divide. By reconstructing these flows and combining them with domestic trade, we obtain a structurally interpretable measure of trade barriers. The resulting estimates show that geopolitical fragmentation generated large and strongly time-varying trade costs that closely track major geopolitical developments.

The paper also relates to work on the Cold War, the Iron Curtain, and the integration of Europe. This literature studies how political division shaped trade across the continent, often focusing on cross-sectional differences or on the post-Cold War adjustment (Eichengreen, 2007a; Nitsch and Wolf, 2013; van Bergeijk, 2015; Beestermöller and Rauch, 2018; Egger et al., 2025). We differ by estimating the evolution of trade barriers during the Cold War itself. This time dimension allows us to connect changes in trade barriers to shifts in East–West relations, rather than measuring only average differences across groups or post-Cold War adjustment. The Cold War therefore provides a historical setting in which political alignment was persistent, while the intensity of confrontation changed substantially over time.

Finally, we build on the structural gravity literature, which measures trade frictions from observed trade flows. Following McCallum (1995), Anderson and van Wincoop (2003), and the empirical gravity framework summarized by Head and Mayer (2014), we estimate trade barriers from bilateral variation after accounting for exporter-year and importer-year factors.

Recent work shows how these specifications can be given a structural interpretation (Fally, 2015; Yotov et al., 2016) and how observed trade flows can be used to recover implied trade costs (Novy, 2013; Jacks et al., 2011). Our setting highlights a practical requirement of this approach: recovering the trade barriers associated with geopolitical fragmentation requires a matrix of bilateral and domestic trade flows. Our data reconstruction completes and corrects this trade matrix in order to improve the estimation of trade costs.

Roadmap

Section 2 describes the Cold War setting and the evolution of geopolitical tensions across blocs. Section 3 presents the construction of the bilateral and domestic trade data. Section 4 introduces the structural gravity framework and estimation strategy, and Section 5 reports the estimated evolution of trade barriers between Eastern and Western Europe during the Cold War. Section 6 examines how geopolitical fragmentation shaped trade relationships beyond the core East–West divide. Section 7 discusses the broader interpretation of the results and their relevance for current debates on geopolitical fragmentation. Section 8 concludes. Additional robustness exercises and welfare calculations are reported in the appendix.

2 Historical Background

2.1 The Cold War as a laboratory for fragmentation

The Cold War provides a unique setting to study large-scale geopolitical fragmentation. During this period, the international system was organized around a division into two competing blocs, which extended beyond political and military rivalry to shape international relations and constrain cross-bloc interactions.

This fragmentation was both global in scope and highly persistent. For several decades, a substantial share of the world economy operated within a bipolar structure in which interactions across blocs were systematically restricted. Compared to more recent episodes, which tend to be partial or short-lived, the Cold War represents a sustained and pervasive form of fragmentation.

The intensity of fragmentation changed over time, with periods of heightened confrontation alternating with phases of partial easing. These changes plausibly affected the costs of trading across blocs, and provide useful variation to study how geopolitical conditions relate to trade.

Although these changes were not fully exogenous, they were driven mainly by geopolitical and strategic considerations rather than by trade-related factors. This makes the Cold War a useful

setting to study changes in trade barriers that are largely orthogonal to standard economic determinants of bilateral trade.

A central challenge is measurement. Standard international trade datasets provide incomplete coverage of bilateral trade flows involving Eastern bloc countries, with particularly large gaps in trade within the bloc. This makes it difficult to construct consistent trade matrices and to measure trade costs, and requires reconstructing missing flows to obtain a complete picture of fragmentation.

2.2 The Iron Curtain and the European division

In practice, we focus on Europe, where geopolitical fragmentation took its most concrete form. The Iron Curtain marked the boundary between Eastern and Western Europe, running across the continent as a fortified border with walls, fences, checkpoints, and tightly controlled crossing points. It reflected a broader political divide between Soviet-aligned governments in the East and market-oriented democracies in the West, as well as a military divide anchored in opposing alliances. Economic exchange was also limited by policy. Governments on both sides imposed restrictions on trade, including export controls, licensing requirements, and limits on technology transfers, alongside constraints on travel, migration, and economic cooperation, which substantially reduced trade between countries on either side.

Europe provides a useful setting to study fragmentation because countries on either side of the Iron Curtain were geographically close and, in some cases, had strong trade links prior to the division. This suggests that there was scope for trade in the absence of barriers, so differences in trade across the divide are more likely to reflect geopolitical constraints rather than geographic factors.

The European division also offers a clear and persistent source of variation. The boundary between the blocs was well defined and remained largely stable over time, with little ambiguity about country alignment. This differs with more diffuse forms of fragmentation in other regions of the world and allows a cleaner comparison between trade within blocs and trade across the divide.

Data considerations further reinforce the focus on Europe. Standard sources provide incomplete coverage of trade involving Eastern bloc countries, with especially large gaps in trade within the bloc, making it difficult to form a consistent picture of trade patterns. While these measurement issues are present in other regions, they are especially acute in Eastern Europe and motivate the data construction used in our analysis.

While our analysis focuses on Europe for identification, we show in Section 6 that similar

patterns emerge in a broader global sample, suggesting that the relationship between geopolitical fragmentation and trade is not region-specific.

The evolution of this division over time provides additional variation that we exploit empirically. We next describe the main geopolitical phases of the Cold War in Europe.

2.3 Major geopolitical phases

The evolution of East–West relations during the Cold War can be described in terms of a set of broad geopolitical phases, each associated with different levels of tension and scope for interaction between the blocs (see also Table A.1 in the appendix). While the territorial division of Europe remained broadly stable after the late 1940s, the intensity of confrontation and the extent of economic and political exchange changed over time.

The first phase corresponds to the formation of the opposing blocs in the late 1940s and early 1950s. The Marshall Plan, implemented from 1948 as a U.S.-funded program for European reconstruction and economic coordination, supported recovery in Western Europe and required institutional cooperation through the Organisation for European Economic Co-operation (OEEC) (Milward, 1984; Eichengreen, 2007b). The creation of NATO in 1949 formalized the security dimension of this alignment. In Eastern Europe, the Soviet Union established political control and introduced centrally planned economies, alongside the creation of the Council for Mutual Economic Assistance (CMEA) in 1949 to organize trade and production among socialist economies (Kaser, 1965). The Berlin Blockade of 1948–49 made the division of Germany explicit, while the Korean War (1950–53) reinforced the global character of the conflict and contributed to tighter security policies, including export controls. By the early 1950s, the separation between the blocs was largely in place.

A second phase begins in the mid-1950s, following Stalin's death in 1953. Changes in Soviet leadership were accompanied by a somewhat less rigid foreign policy. Diplomatic initiatives such as the Geneva Summit of 1955—an early effort to ease tensions among the major powers—and the Austrian State Treaty, which restored Austria's sovereignty under a commitment to neutrality, indicate a partial easing of tensions in European relations (Gaddis, 2005; Leffler, 2007). At the same time, tensions remained. The suppression of the Hungarian uprising in 1956 showed the limits of political autonomy within the Eastern bloc. Disputes over Berlin continued, and competition extended beyond Europe, including the Cuban Revolution and developments in strategic technology after the launch of Sputnik in 1957. These tensions culminated in the construction of the Berlin Wall in 1961 and the Cuban Missile Crisis in 1962..

The third phase corresponds to the period of *Détente*, which developed from the late 1960s. In Europe, *Ostpolitik* under Willy Brandt led to a series of agreements between West Germany and

Eastern European countries, including the Moscow Treaty of 1970, which recognized postwar borders and renounced the use of force, and the Basic Treaty of 1972, which normalized relations between East and West Germany (Gaddis, 2005). Relations between the United States and the Soviet Union also became more structured, as negotiations on nuclear arms produced the SALT I agreement and the Anti-Ballistic Missile Treaty in 1972, both aimed at limiting strategic weapons systems. The Helsinki Final Act of 1975, signed within the framework of the Conference on Security and Cooperation in Europe, set out principles on territorial integrity, economic cooperation, and human rights, providing a basis for continued engagement across blocs. Political rivalry remained, but the environment became more predictable, and some areas of economic exchange expanded, including agricultural trade and selected industrial sectors. This phase ended with the Soviet invasion of Afghanistan in 1979.

The late 1970s and early 1980s mark a fourth phase of renewed tension. The Soviet intervention in Afghanistan led to a deterioration in relations and to the suspension of several forms of cooperation. Military competition intensified, including the deployment of intermediate-range nuclear forces in Europe and the NATO dual-track decision, which combined arms deployment with an offer of negotiations. The election of Ronald Reagan in 1981 coincided with a shift in U.S. policy, with a military build-up, tighter restrictions on technology transfers, and a more confrontational stance toward the Soviet Union. In his electoral campaign, Reagan had argued that the textitDétente had allowed the Soviet Union to gain advantages without sufficient reciprocity, and this view shaped policy choices in the early 1980s (Gaddis, 2005). The Strategic Defense Initiative, announced in 1983, proposed a missile defense system intended to intercept nuclear weapons, and added to tensions by challenging the existing framework of nuclear deterrence.

A fifth phase begins in the mid-1980s with a gradual easing of tensions. The appointment of Mikhail Gorbachev in 1985 led to changes in both domestic and foreign policy. His reform agenda included *perestroika* and *glasnost*, aimed at restructuring the Soviet economy through limited market mechanisms and liberalizing political and public life. These reforms were accompanied by a shift toward negotiation with the United States. A series of summits resumed dialogue, beginning with the Geneva meeting in 1985, and led to arms control agreements, including the Intermediate-Range Nuclear Forces (INF) Treaty of 1987, which eliminated an entire class of nuclear missiles. During this period, the Soviet Union also reduced its willingness to intervene in Eastern Europe (Brown, 2009).

The final phase corresponds to the collapse of the Eastern bloc. The fall of the Berlin Wall in 1989 marked the end of the division of Europe, followed by German reunification in 1990 and the dissolution of the Soviet Union in 1991. Political change was rapid, but economic adjustment took longer. Trade patterns, production structures, and institutional arrangements

shaped during the Cold War did not disappear immediately, and their effects persisted beyond its end (Roland, 2000).

3 Data

3.1 Main sample and bloc definitions

Countries are classified into Eastern and Western blocs based on their participation in the economic and military institutions that defined the Cold War divide, as documented in Tables A.2–A.4 in the appendix.

In our baseline analysis we restrict attention to European countries to align the empirical design with the core geography of Cold War geopolitical fragmentation, where the division between the blocs was most clearly defined. This allows us to compare geographically proximate countries with a history of economic links, helping to isolate trade frictions associated with geopolitical division. Our data construction improves the measurement of bilateral trade flows for Eastern European economies and makes it possible to implement this comparison in a consistent way.

The division between Eastern and Western Europe during the early Cold War was closely linked to participation in the Marshall Plan and the Council for Mutual Economic Assistance. Although the Marshall Plan was formally open to all European countries, the Soviet Union opposed participation and exerted strong political pressure on countries within its sphere of influence to opt out. For example, Poland and Czechoslovakia initially considered joining, but ultimately withdrew under Soviet pressure. This episode helped define the economic separation of Europe: Western European countries entered a U.S.-led system of cooperation and reconstruction through the OEEC, while Eastern European countries became integrated into a separate Soviet-led system. This historically grounded division provides a natural basis for defining geopolitical blocs in our empirical analysis.

We define the Eastern bloc as the set of European countries belonging to the Soviet sphere, namely the members of the Council for Mutual Economic Assistance and the Warsaw Pact: the Soviet Union, East Germany, Poland, Czechoslovakia, Hungary, Romania, and Bulgaria. The Western bloc is defined as the group of European countries that participated in the Marshall Plan and were aligned with the Western security alliance. Our baseline Western sample therefore includes West Germany, France, Italy, Belgium, the Netherlands, Luxembourg, the United Kingdom, Denmark, Norway, Portugal, and Greece. We exclude countries that were not consistently aligned with either bloc throughout the Cold War, such as Spain, Austria, Finland, and Yugoslavia, to maintain a clear distinction between politically aligned groups. Iceland is also excluded. Despite its participation in the Marshall Plan institutions and NATO,

its geographic isolation and very limited trade with the rest of Europe provide little variation for identifying trade costs.

3.2 The limits of standard trade data

Standard international trade datasets provide an incomplete and uneven picture of trade during the Cold War, especially for countries in the Eastern bloc. The IMF's Direction of Trade Statistics (DOTS), the main source of historical bilateral trade data for this period, relies on official reporting by national statistical authorities. Since most Eastern European countries and the Soviet Union were not IMF members during much of this period, their trade flows are only partially captured.

Figure 1 reports the number of non-missing bilateral trade flows by directed country pair. Coverage of trade involving Eastern bloc countries is highly uneven, with many flows—especially within the Eastern bloc–entirely missing. When observations are available, they often come from mirror statistics reported by trading partners rather than direct reporting. Appendix B.1 presents additional evidence on the reporting structure behind these patterns.

Importantly, these gaps do not simply reflect a lack of underlying data. In several cases, bilateral flows appear in the DOTS data for limited periods and then disappear. For example, in the mid-1950s the dataset briefly includes a broader set of flows involving Eastern European countries, including economically important transactions that are otherwise unreported. This pattern suggests that the limitations of standard datasets arise not only from data availability, but also from incomplete and inconsistent integration of existing sources.

A clear example is trade between East and West Germany. This was one of the largest and most economically significant trade relationships across the Cold War divide, yet it is largely absent from standard international datasets. The omission reflects institutional classification: the two German states did not recognize each other as foreign countries and recorded these transactions as *Innerdeutscher Handel* (intra-German trade), outside the framework of international trade statistics. As a result, these flows were not incorporated into datasets such as DOTS, despite being recorded in official sources.

Taken together, these features imply that standard trade data provide an incomplete picture of trade during the Cold War. They systematically understate trade involving Eastern bloc countries and omit important flows both within the East and across the East–West divide. As a result, taken at face value, standard datasets prevent the construction of a complete and internally consistent bilateral trade matrix and the implementation of a structural gravity framework for measuring trade frictions.

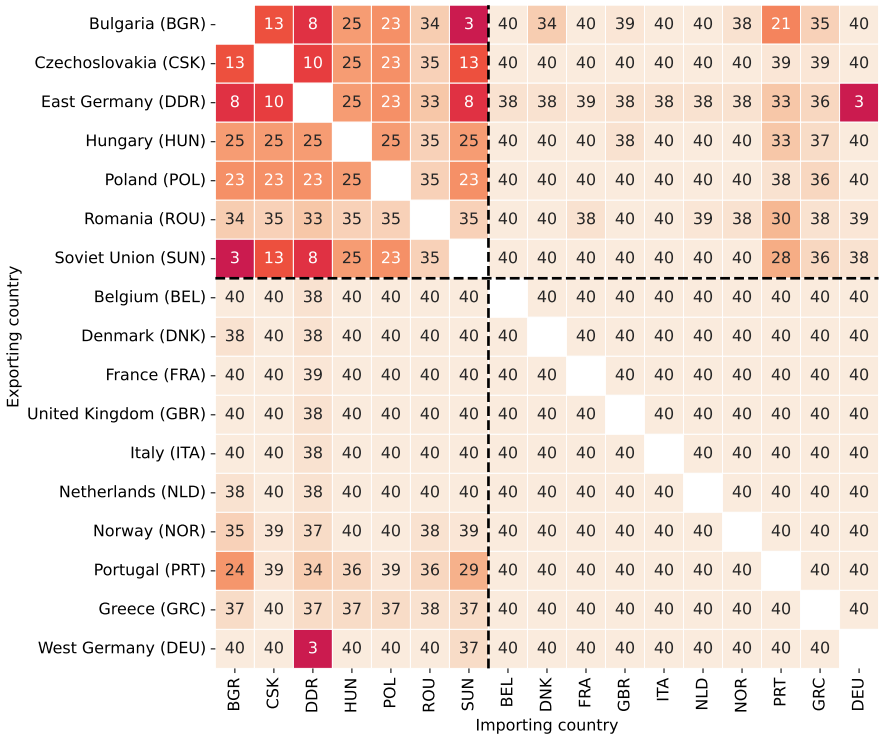


Figure 1: Coverage of bilateral trade flows in DOTS (1950–1989)

Notes: Each cell reports the number of years with non-missing bilateral trade flows for a directed country pair in the DOTS database. Exporters are shown in rows and importers in columns. A dashed line separates Eastern and Western bloc countries.

3.3 Primary sources and bilateral trade reconstruction

To address these limitations, we construct a new dataset of bilateral trade flows using primary historical sources. Detailed descriptions of the data sources, transformations, and construction procedures are provided in Appendix B.

During the Cold War, countries in the Eastern bloc regularly published detailed information on their foreign trade in national statistical yearbooks and specialized reports. While these publications contain substantial information on bilateral trade flows, they were not systematically incorporated into international datasets such as DOTS.

We collect bilateral trade data for the main Eastern European economies—including Bulgaria, Czechoslovakia, East Germany, Hungary, Poland, and the Soviet Union—from their respective statistical publications. These sources report trade flows by partner country on a consistent annual basis, allowing us to recover large parts of the bilateral trade network that are missing in standard data. Importantly, they also provide direct observations of trade flows that are only indirectly measured in standard datasets through mirror statistics. This improves the measurement of bilateral trade not only for exchanges within blocs, but also for trade between Eastern and Western countries.

A number of cases require special treatment. Trade between East and West Germany, one of the largest flows across the Cold War divide, was recorded outside the standard framework of international trade statistics. We draw on three complementary sources: official statistics from West Germany, official statistics from East Germany, and a post-reunification reconstruction produced by the Statistisches Bundesamt, which combines information from both sides. This allows us to recover a consistent series over the full sample period.

We then combine the information from these primary sources with existing data from DOTS to construct a comprehensive bilateral trade matrix covering both Eastern and Western European countries. This approach allows us to retain the broad coverage and comparability of standard datasets while substantially improving the measurement of trade involving the Eastern bloc. In particular, it enables us to fill in missing flows within the Eastern bloc and to replace mirror-based observations with directly reported data wherever possible.

The resulting dataset provides a more complete and consistent representation of trade patterns during the Cold War, forming the basis for our empirical analysis. Taken together, these steps allow us to replace a large share of missing or mirror-based observations in standard datasets with directly reported bilateral flows. The resulting dataset improves both the coverage and the measurement of trade involving Eastern bloc countries, while remaining fully comparable to standard sources.

3.4 Processing data using the DOTS methodology

Our objective is to ensure that differences between our data and standard datasets reflect improvements in coverage rather than differences in measurement. To ensure comparability with standard international datasets, we process the data from primary sources following the methodology used in the IMF's Direction of Trade Statistics (DOTS). Conceptually, we treat the collected data as if they had been reported within the DOTS system and apply the same transformations that the IMF would have performed.

First, we account for revisions in the original sources by updating reported values to their latest available version when discrepancies arise across successive publications. This ensures internal consistency within each country's statistical series.

Second, we convert all trade values into U.S. dollars using official exchange rates. We take these exchange rates from the United Nations, which computed and published conversion factors designed specifically for expressing trade flows in U.S. dollars. This step harmonizes values across countries and over time.

Third, we address differences between f.o.b. (free on board) and c.i.f. (cost, insurance, and freight) valuation. Following the DOTS methodology, we adjust import values reported on an f.o.b. basis by applying the standard conversion used in the DOTS framework to approximate their c.i.f. equivalent. For the period under study, this implies a 10 percent adjustment, as prescribed in the DOTS methodology.

Applying these steps yields a dataset that is directly comparable to DOTS while incorporating a substantially richer set of underlying observations. As a result, any remaining differences relative to standard sources mainly reflect improved coverage rather than differences in construction.

3.5 Validation

We assess the consistency of the reconstructed trade data using external sources. In particular, we compare our series with declassified CIA trade data for the period 1960–1970, where coverage overlaps. The two sources align closely in both levels and trends, indicating that the harmonization procedure yields data comparable to standard international statistics.

We also verify internal consistency by comparing importer- and exporter-reported flows when both are available. Discrepancies are similar in magnitude to those observed in standard datasets such as DOTS, suggesting that the reconstructed series are not subject to systematic distortions.

3.6 Constructing domestic trade

To interpret international trade frictions relative to trading at home, we complement the bilateral trade data with estimates of domestic trade. Rather than developing a new methodology, we follow the approach of Head and Mayer (2021), who construct domestic trade in goods as the difference between goods production and total exports, building on earlier work that applied this method to manufacturing (Baier et al., 2019).

The starting point is a country-year panel of aggregate goods production. Following Head and Mayer (2021), this is recovered from GDP and sectoral information on output and value added, using interpolated output-to-GDP ratios where direct production data are unavailable. Domestic trade is then obtained as goods production minus aggregate exports. This procedure provides a standard way to recover the domestic benchmark needed to express international trade frictions as border effects.

We implement the Head and Mayer (2021) procedure in Python and verify that our translation reproduces their original dataset. We then adapt the construction in a limited number of ways to fit our historical setting. First, we retain historical entities that are excluded from the standard dataset, most importantly Czechoslovakia. Second, we extend the extrapolation of the output-to-GDP ratio used in the construction of goods production to the early years of our sample for Eastern European countries, so as to obtain coverage back to 1950. Third, we eliminate discontinuities in GDP series for historical countries by relying on consistent historical data and extending coverage using growth rates from alternative sources. In addition, we replace mismeasured trade data for East Germany with series constructed from primary sources. Appendix B.5 provides more detail on the data issues and adjustments made for constructing domestic trade. For East Germany, where no initial observation is available from which to extrapolate the output-to-GDP ratio, we impute this ratio using the contemporaneous average of the other Eastern bloc economies in our sample.

These adjustments ensure that the standard domestic trade construction can be applied in a historical Cold War setting.

4 Empirical Strategy

Our empirical design focuses on a set of European countries assigned to one of the two Cold War blocs.[1]

[1]For countries that break up after the end of the Cold War, we use their European successor states: the Czech Republic and Slovakia replace Czechoslovakia and Estonia, Latvia, Lithuania, Moldova, Ukraine, and Russia replace the Soviet Union.

The estimation sample excludes European countries that were not aligned with either bloc. This restriction is part of the identification strategy, rather than a simple sample selection choice. It focuses the analysis on the core of the Cold War division and on the part of the trade matrix where our new data are most informative. Within this sample, each bilateral observation falls into one of four mutually exclusive categories: East–East, East–West, West–East, or West–West.

We now examine which comparisons across these groups can be identified once exporter-year and importer-year fixed effects are included in the regression.

4.1 Initial specification: international trade flows

We begin with international trade flows only, excluding domestic observations. Let X_{ijt} denote exports from country i to country j in year t, with $i \neq j$. We estimate the following structural gravity equation:

$$X_{ijt} = \exp\Big(\gamma_t^{CB} CB_{ij} + \phi_{it} + \psi_{jt} + z_{ij}'\beta\Big) + \varepsilon_{ijt}, \tag{1}$$

where ϕ_{it} and ψ_{jt} denote exporter-year and importer-year fixed effects, and z_{ij} is a vector of standard bilateral gravity controls (distance and contiguity).

The variable CB_{ij} is an indicator for cross-bloc trade:

$$CB_{ij} \equiv \mathbf{1}\{i \in East,\ j \in West\} + \mathbf{1}\{i \in West,\ j \in East\}.$$

In the bloc-restricted European sample, the international trade matrix contains four directional bloc-pair types: East–East, East–West, West–East, and West–West. However, once exporter-year and importer-year fixed effects are included, these four categories do not generate four independent sources of variation. The fixed effects absorb all exporter-side and importer-side components, including any year-specific differences between Eastern and Western countries. What remains identified is a single bilateral contrast: whether a trade flow crosses the bloc boundary or stays within a bloc. Put differently, the international-only specification identifies cross-bloc trade relative to within-bloc trade, where the latter is not separately decomposed into East–East and West–West components. A formal characterization of this result is provided in Appendix C.

This identification result also implies that directional asymmetries in cross-bloc trade are not separately identified. East-to-West and West-to-East flows can differ in the data, but conditional on exporter-year and importer-year fixed effects that directional difference is not separately estimable in the bloc-restricted two-group design. The coefficient γ_t^{CB} should therefore be interpreted as the symmetric cross-bloc trade wedge relative to within-bloc international trade.

This baseline specification is attractive because it relies exclusively on directly observed bilateral trade flows. It provides a transparent measure of fragmentation using the strongest part of the data, but its identifying variation is necessarily one-dimensional. Its main drawback is that the cross-bloc trade wedge cannot be separated from the evolution of within-bloc trade frictions.

4.2 Augmented specification: including domestic trade

We next augment the analysis by incorporating domestic trade flows. Let X_{iit} denote domestic trade in country i at time t. We estimate

$$X_{ijt} = \exp\Big(\gamma_t^{CB} CB_{ij} + \gamma_t^{EE} EE_{ij} + \gamma_t^{WW} WW_{ij} + \phi_{it} + \psi_{jt} + z'_{ij}\beta\Big) + \varepsilon_{ijt}, \tag{2}$$

where the sample now includes both international and domestic observations, and domestic trade ($i = j$) is the omitted category. Bloc membership is time-invariant in our setting, so we omit time subscripts on the bloc indicators.

The variables EE_{ij} and WW_{ij} are defined as

$$EE_{ij} \equiv \mathbf{1}\{i \in East,\ j \in East,\ i \neq j\}, \qquad WW_{ij} \equiv \mathbf{1}\{i \in West,\ j \in West,\ i \neq j\}.$$

Adding domestic trade changes the dimensionality of the identification problem. In the international-only specification, the bilateral variation generated by the bloc structure collapses to a single contrast once exporter-year and importer-year fixed effects are included. Domestic trade introduces an additional benchmark that is not spanned by exporter-year and importer-year fixed effects: for domestic observations, $CB_{ii} = EE_{ii} = WW_{ii} = 0$. Intuitively, domestic observations lie on the diagonal of the trade matrix and therefore introduce variation that cannot be replicated by exporter-side or importer-side effects alone. As a result, the model can separately identify cross-bloc trade, East–East trade, and West–West trade relative to trading at home. Appendix C shows that, with domestic observations included, the relevant bloc-pair indicators are linearly independent conditional on exporter-year and importer-year fixed effects.

The coefficients in (2) therefore admit a border-effect interpretation relative to domestic trade. The coefficient γ_t^{CB} measures the trade wedge associated with crossing the bloc boundary relative to domestic trade. The coefficients γ_t^{EE} and γ_t^{WW} measure the trade wedges associated with within-bloc international trade in the East and West, respectively, again relative to domestic trade. As in the international-only specification, East-to-West and West-to-East flows are not separately identified once exporter-year and importer-year fixed effects are included, so the cross-bloc coefficient captures their symmetric component.

Including domestic trade introduces an additional normalization that is not present in the international-only specification. In the augmented model, domestic observations form the omitted category, so all bilateral trade flows are evaluated relative to trading at home. After conditioning on exporter-year and importer-year fixed effects, this treats domestic trade as a common benchmark across countries and over time: the fixed effects absorb country-year differences in overall trade levels, but the specification does not allow for an additional domestic-specific component that varies across countries or over time beyond what those effects capture. Under this normalization, the coefficient on cross-bloc trade can be interpreted as the trade wedge associated with crossing the bloc boundary relative to domestic trade. While this normalization is natural in a structural gravity framework, it relies on the assumption that domestic trade provides a stable benchmark across countries and over time. In practice, it is difficult to assess whether the ease of trading domestically evolved differently across the two blocs in ways not captured by the fixed effects. If such differences exist, the estimated cross-bloc coefficient will reflect not only changes in trade across blocs, but also shifts in the domestic benchmark.

4.3 Interpretation and complementarity of the two specifications

The two specifications serve different but complementary purposes.

The international-only specification provides the cleanest estimate of fragmentation in the bloc-restricted European sample. Because it relies solely on observed bilateral trade flows, it minimizes dependence on auxiliary assumptions and isolates the one bilateral contrast that is identified in a two-bloc design with exporter-year and importer-year fixed effects: cross-bloc trade relative to within-bloc trade.

At the same time, the international-only design cannot separately identify the influence of within-bloc trade. The domestic-trade specification addresses this limitation. Although domestic trade is not directly observed in our historical setting and must be constructed using auxiliary data and standard procedures from the literature, its inclusion yields substantial gains in identification. It allows us to distinguish whether the estimated cross-bloc wedge reflects unusually low trade across blocs, unusually high trade within the West, unusual patterns of trade within the East, or some combination of these forces.

We therefore view the two specifications as complementary. The first provides a parsimonious and data-transparent measure of geopolitical fragmentation based entirely on observed international flows. The second introduces a richer benchmark that makes it possible to decompose trade patterns across bloc-pair types and to express them as border effects relative to domestic trade. Taken together, they allow us to separate what is identified from observed bilateral variation alone from what can be learned once domestic trade is introduced as an additional benchmark.

5 Trade Costs and Geopolitical Fragmentation

5.1 International trade flows

Before introducing domestic trade, we first examine what can be learned from observed international trade flows alone. This specification provides a transparent measure of cross-bloc trade frictions because it relies exclusively on directly observed bilateral trade data. However, the estimated coefficient captures cross-bloc trade relative to within-bloc international trade, pooling East–East and West–West flows into a common benchmark. As a result, the specification identifies whether trade across the bloc boundary is systematically more difficult than trade within blocs, but it does not separately isolate the cross-bloc component of trade barriers from broader changes in economic integration within blocs.

Cross-bloc trade frictions estimated from international flows alone are large throughout the Cold War. Figure 2 reports the implied ad valorem tariff equivalent of cross-bloc trade under this specification, using a conventional trade elasticity of 5 (Head and Mayer, 2014). The estimated barriers are equivalent to a tariff of roughly 50 percent relative to within-bloc international trade, indicating substantial frictions between Eastern and Western Europe. At the same time, the estimated barriers appear relatively persistent over time, with limited evidence of a pronounced secular decline. This pattern reflects an important limitation of the international-only specification: because the benchmark is within-bloc international trade, changes in cross-bloc trade frictions are not separately identified from broader changes in economic integration within blocs.

The comparison with standard international trade data highlights the importance of reconstructing missing bilateral flows. Figure 2 also reports estimates obtained using the original IMF Direction of Trade Statistics (DOTS). While the two series are broadly similar early in the sample, the DOTS-based estimates decline sharply from the 1960s onward, in contrast to the relatively stable pattern obtained using the reconstructed data. This difference reflects the incomplete coverage of trade involving socialist economies in the standard data, particularly the omission of economically important intra-bloc trade flows. As a result, standard datasets yield a misleading picture of the evolution of trade integration across the Cold War divide.

To isolate the cross-bloc component of trade barriers, we next introduce domestic trade as an explicit benchmark.

5.2 Cross-bloc trade costs relative to domestic trade

Introducing domestic trade changes the interpretation of the estimated trade barriers by providing a benchmark relative to trading within national borders. Relative to the international-

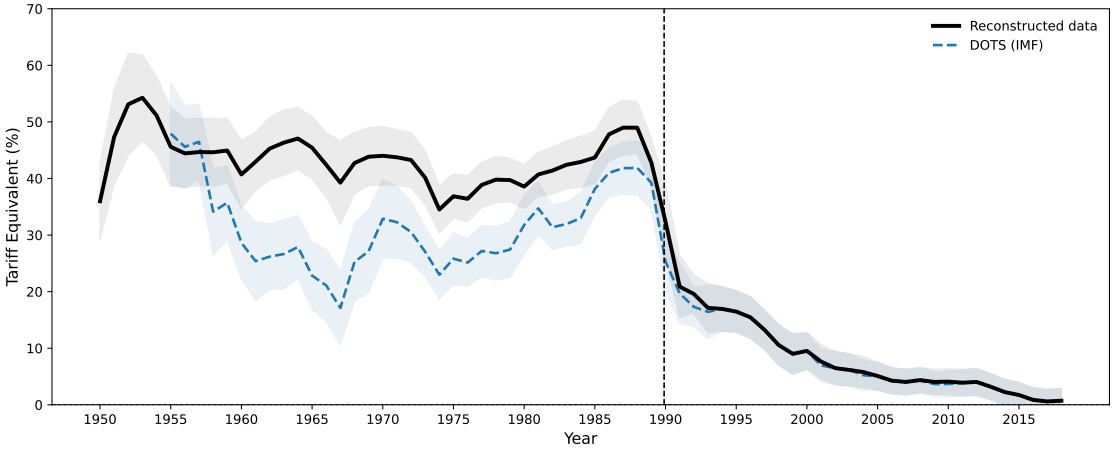


Figure 2: Cross-bloc trade costs based on international trade flows

Notes: The figure shows the estimated tariff equivalent of trade between countries located in different Cold War blocs (East–West), expressed in percentage points. The solid line reports estimates based on the reconstructed dataset, while the dashed line reports estimates based on the original IMF Direction of Trade Statistics (DOTS). Both series are obtained from specification (1) using only international trade flows. The tariff equivalent is obtained by converting the estimated coefficients into ad valorem trade costs using a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t^{CB}/5) - 1]$. Shaded areas represent 95% confidence intervals, constructed using the delta method. A vertical dashed line marks the fall of the Berlin Wall.

only specification, this additional benchmark isolates the cross-bloc component of trade frictions from broader changes in within-bloc integration. The resulting estimates can therefore be interpreted as border effects relative to domestic trade and are directly comparable to standard measures of trade costs in the gravity literature.

Once domestic trade is introduced as a benchmark, cross-bloc trade barriers display a clear and economically significant evolution over time. Figure 3 reports the implied tariff equivalent of cross-bloc trade under this specification. The estimated barriers are large throughout the sample, indicating substantial frictions between Eastern and Western Europe relative to domestic trade. More importantly, the barriers closely track major geopolitical developments over the Cold War. They decline during periods of easing East–West tensions, rise again during renewed geopolitical confrontation in the early 1980s, and fall sharply during the final phase of the Cold War. These patterns show that trade barriers associated with geopolitical fragmentation varied closely with geopolitical developments, rather than remaining constant over time.

The contrast with the international-only specification reflects the importance of within-bloc integration during the Cold War. Relative to within-bloc international trade, cross-bloc barriers appear comparatively stable over time. Once domestic trade is introduced as a benchmark, however, the evolution of cross-bloc trade barriers becomes much more pronounced. This difference indicates that integration within blocs deepened substantially over the Cold War period, particularly within Western Europe.

5.3 Geopolitical fragmentation and European integration

During the Cold War, patterns of economic integration in Europe evolved both across blocs and within blocs. Figure 4 compares the evolution of trade costs within the Eastern bloc, within the Western bloc, and across the East–West divide.

Trade costs within both blocs decline substantially over time, while cross-bloc trade barriers remain persistently much higher throughout the sample. Europe therefore became more economically integrated within blocs while remaining fragmented across the geopolitical divide.

The decline in within-Western-bloc trade costs is relatively smooth and persistent, consistent with the gradual deepening of European economic integration in the West.

Integration within the Eastern bloc also deepens over time, although less smoothly. Intra-Eastern-bloc trade frictions rise temporarily during the early 1980s, consistent with the broader economic and political disruptions affecting socialist economies during this period. Despite these fluctuations, the longer-run pattern remains one of increasing integration within the Eastern bloc.

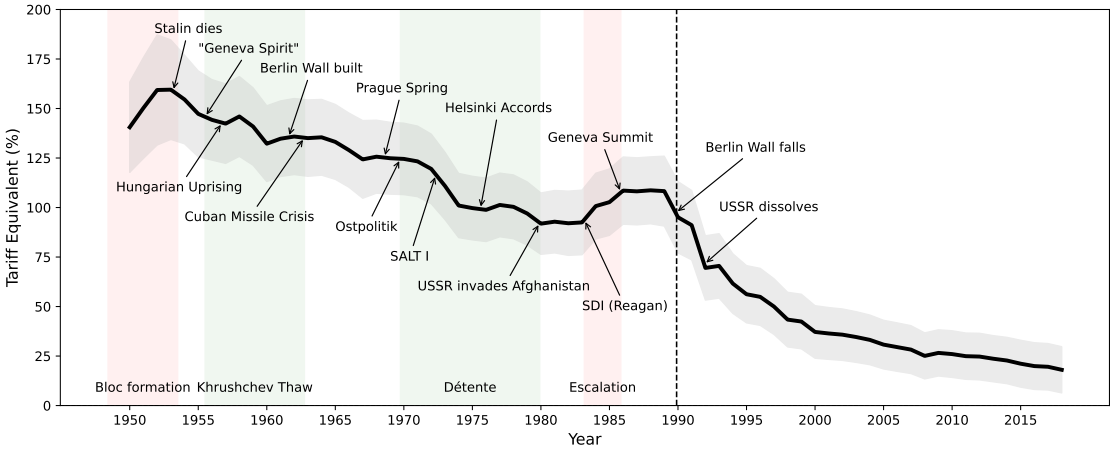


Figure 3: Cross-bloc trade costs during the Cold War

Notes: The figure shows the estimated tariff equivalent of trade between countries located in different Cold War blocs (East–West), expressed in percentage points. The estimates are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is obtained by transforming the estimated coefficient $\hat{\gamma}_t^{CB}$ into an ad valorem trade cost using the standard gravity mapping with a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t^{CB}/5) - 1]$. The shaded area represents the 95% confidence interval, constructed using the delta method. Annotations denote selected geopolitical events, and shaded bands indicate the main Cold War phases described in Section 2 and summarized in Table A.1.

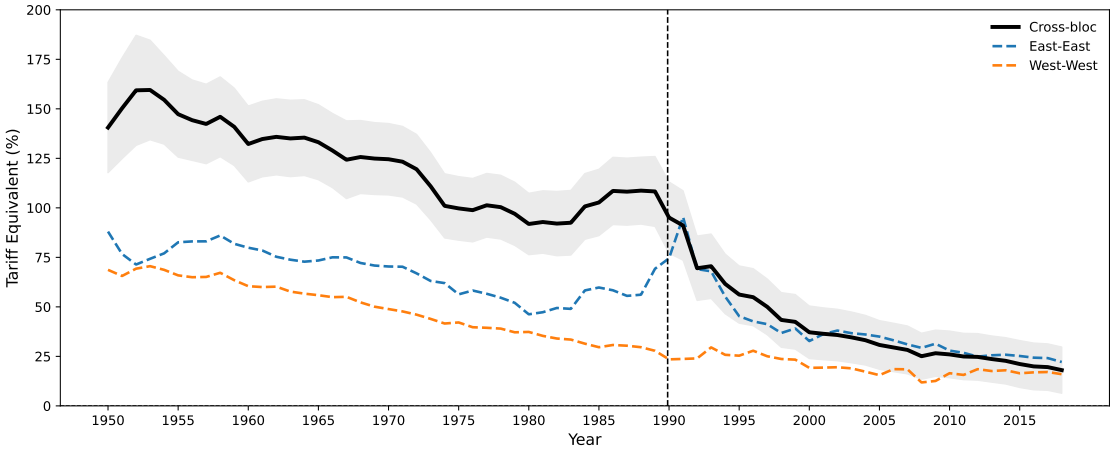


Figure 4: Cross-bloc and intra-bloc trade costs during the Cold War

Notes: The figure shows the estimated tariff equivalent of trade between countries located in different Cold War blocs (East–West) and in the same Cold War bloc (East–East and West–West), expressed in percentage points. The estimates are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is obtained by transforming the estimated coefficients into an ad valorem trade cost using the standard gravity mapping with a trade elasticity of 5. The shaded area represents the 95% confidence interval for the cross-bloc estimate, constructed using the delta method. A vertical dashed line marks the fall of the Berlin Wall.

5.4 Validation and measurement issues

We next assess whether the estimated evolution of cross-bloc trade barriers is sensitive to the two main measurement concerns in our setting: the construction of domestic trade and the institutional organization of trade within the socialist bloc.

Domestic trade measurement

Domestic trade is not directly observed in our setting and must be constructed using auxiliary data on output and sectoral composition. To assess whether the results depend on these measurement choices, we consider alternative specifications that change both the evolution of the tradables share and the method used for the construction of output. We consider a less variable share of tradeables, a smoother series for output, and an alternative source for GDP data. Across specifications, the estimated evolution of cross-bloc trade barriers remains highly stable, with only modest differences in levels. Details and additional results are reported in Appendix D.2.

Valuation and pricing distortions in East–East trade

A separate concern arises from the institutional organization of trade within the socialist bloc. Trade within the Council for Mutual Economic Assistance (CMEA) operated under a distinct pricing and accounting system that differed substantially from market-based trade. East–East transactions were denominated in non-convertible transferable rubles and often valued using administratively determined prices rather than contemporaneous world prices. As a result, measured trade flows within the socialist bloc may partly reflect accounting conventions and redistribution mechanisms rather than directly comparable market valuations.

To assess whether these features affect the results, we re-estimate the model excluding all East–East trade flows while retaining domestic trade for identification. This removes the CMEA valuation regime and restricts the analysis to trade conducted in convertible currencies and priced at market conditions. The estimated cross-bloc trade barrier remains remarkably stable relative to the baseline, although standard errors increase slightly. This indicates that intra-CMEA trade does not drive the main findings. Instead, East–East flows primarily contribute to estimation precision, while the identification of the cross-bloc coefficient relies on comparisons between cross-bloc and market-based trade flows. Details and additional results are reported in Appendix D.3.

Additional evidence

Additional robustness exercises reported in Appendix D consider bilateral heterogeneity, alternative bloc definitions, directional asymmetries in cross-bloc trade, different treatments of missing trade flows, and extensions to broader global samples. We also construct bilateral trade cost measures using the symmetric transformation of Head and Ries (2001). Across all exercises, the main findings remain robust.

Taken together, these exercises indicate that the estimated evolution of cross-bloc trade barriers is not driven by the construction of domestic trade or by institutional features specific to intra-socialist trade flows.

6 Geopolitical fragmentation and the structure of trade integration

The previous section establishes that cross-bloc trade costs during the Cold War were large, time-varying, and closely aligned with geopolitical developments. We now examine how these trade frictions shaped the structure of trade costs across partners, focusing on the emergence of alignment patterns and the role of strategic behavior in shaping trade relationships.

6.1 Adjustment margins under geopolitical fragmentation

The estimated cross-bloc trade barriers capture the overall degree of economic separation between Eastern and Western Europe. But geopolitical fragmentation also reshaped the structure of trade integration across countries. Rather than affecting all trade relationships uniformly, fragmentation altered trade costs along specific margins: through selective integration across relationships with especially dense historical and economic ties, through systematic alignment outside formal blocs, and through deeper integration within politically aligned groups.

The following subsections examine these margins in turn. Together, they show that geopolitical fragmentation changed not only the level of trade barriers across blocs, but also the broader organization of trade relationships within Europe.

6.2 Selective integration across the Iron Curtain: Germany

The evolution of cross-bloc trade costs need not reflect uniform changes across all country pairs. In a fragmented geopolitical environment, reductions in trade barriers may be concentrated in

relationships with especially dense historical, social, and economic ties. In Europe, the most important such relationship was trade between East and West Germany.

Trade between the two German states was both economically important and institutionally distinctive. Despite the political division of Germany, substantial economic linkages persisted across the border throughout the Cold War. These flows were officially classified as *Innerdeutscher Handel* (intra-German trade) and therefore treated outside standard international trade statistics. Economic relations between the two Germanies also evolved closely with the political relationship between the two states. Following the construction of the Berlin Wall in 1961, cross-border interactions became more restricted. A gradual normalization began only at the end of the 1960s with the launch of *Ostpolitik* under Chancellor Willy Brandt and the subsequent normalization agreements of the early 1970s. These developments facilitated cross-border movement, payments, and commercial exchange, contributing to a sustained expansion of economic interactions between the two Germanies during the 1970s.[2]

This setting provides a natural way to assess whether the decline in estimated cross-bloc trade barriers during the 1970s reflected broad-based reductions in frictions across all country pairs or instead proceeded through a limited number of economically important bilateral relationships. To evaluate the role of the German relationship, we re-estimate the model excluding both East and West Germany. Figure 5 compares the resulting estimates to the baseline.

Removing East and West Germany changes the time profile of the estimated trade barriers between Eastern and Western Europe. The decline from the early 1970s becomes considerably weaker, and the estimated barriers remain higher during the period of easing East–West tensions and thereafter. This exercise qualifies the aggregate pattern: the reduction in estimated East–West trade barriers during the 1970s was not evenly distributed across country pairs. The German relationship accounts for an important share of the decline in trade barriers, consistent with the role of pre-existing economic links, institutional arrangements, and political normalization in making intra-German trade especially responsive during Détente. The aggregate estimate should therefore be read as reflecting both broader easing across the divide and the exceptional role of intra-German trade.

6.3 Geoeconomic alignment outside formal blocs

Countries outside the formal East–West division provide a useful setting to examine whether geopolitical fragmentation extended beyond formal alliances. In principle, neutral countries could

[2]Key milestones include the signing of the *Basic Treaty* (*Grundlagenvertrag*) in 1972, which established formal relations between West and East Germany, and their mutual recognition and admission to the United Nations in 1973. The intellectual rationale underlying increased economic exchange was closely linked to the logic of *Wandel durch Handel* and Egon Bahr's idea of *Wandel durch Annäherung*, whereby economic engagement was viewed as an instrument of political rapprochement (Bahr, 1990; Blumenau, 2022).

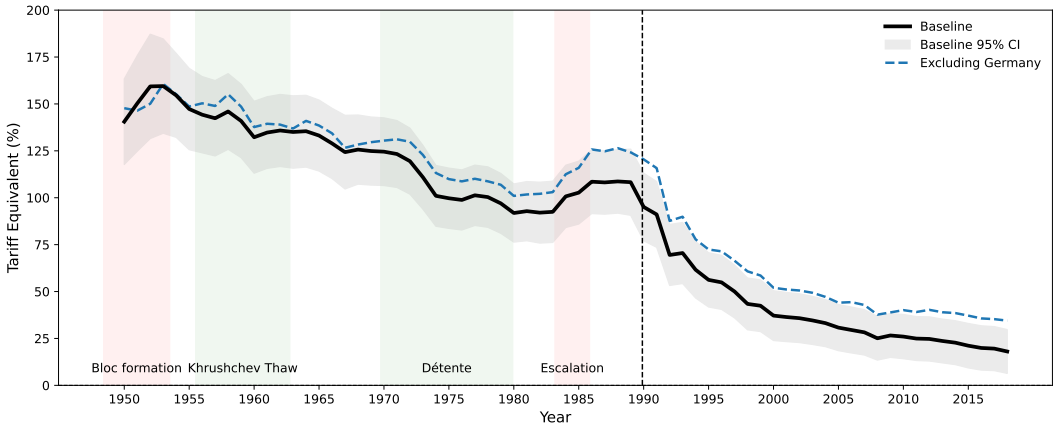


Figure 5: Cross-bloc trade costs excluding East and West Germany

Notes: The figure shows the estimated tariff equivalent of trade between countries located in different Cold War blocs (East–West), expressed in percentage points. The baseline series is compared to an alternative specification that excludes East and West Germany. Estimates are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is computed as $100 \times [\exp(-\hat{\gamma}_t^{CB}/5) - 1]$, using a trade elasticity of 5. The shaded area represents a 95% confidence interval for the baseline estimate, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2, and the vertical dashed line marks the fall of the Berlin Wall.

maintain relatively symmetric economic relationships with both blocs. In practice, differences in their trade costs reveal whether fragmentation generated systematic patterns of geoeconomic alignment even in the absence of formal political affiliation.

To examine this margin, we extend the baseline specification to include a set of European neutral countries (Austria, Switzerland, Sweden, and Ireland) and estimate trade costs between these countries and each bloc. Figure 6 reports the resulting estimates.

Two findings stand out. First, neutral countries faced substantially lower trade costs when trading with the Western bloc than with the Eastern bloc. Trade costs with the West were relatively low and evolved similarly to trade costs within the Western bloc, while trade costs with the East remained substantially higher throughout the sample. Despite their formal neutrality, these countries were therefore much more closely integrated with Western Europe economically.

Second, trade costs between neutral countries and the Eastern bloc closely tracked the evolution of direct East–West trade barriers. The estimated barriers decline during periods of easing geopolitical tensions and rise again during periods of renewed confrontation. This close co-movement suggests that neutral countries were exposed to the same time-varying geopolitical frictions that shaped trade across the broader East–West divide.

Taken together, these findings indicate that geopolitical fragmentation influenced the organiza-

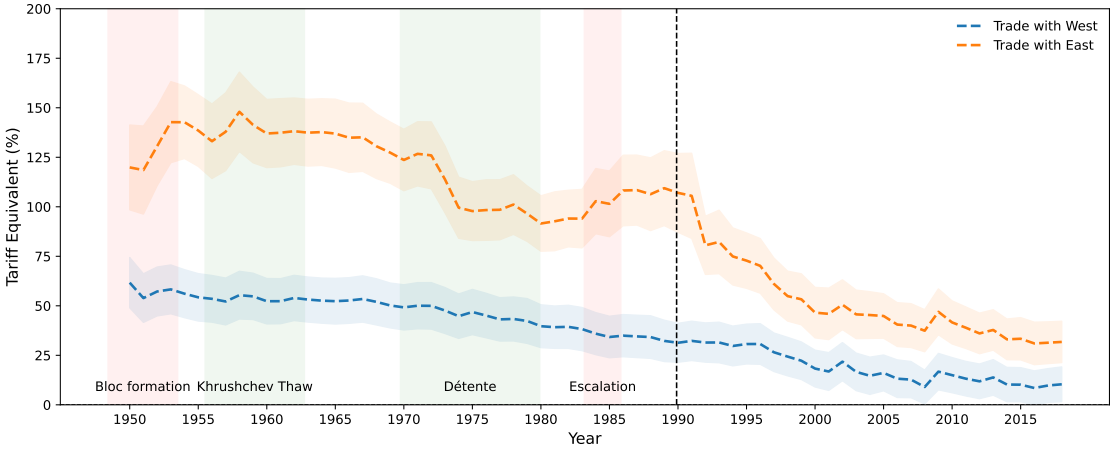


Figure 6: Trade costs for neutral countries

Notes: The figure shows the estimated tariff equivalent of trade between neutral countries and the Eastern and Western blocs, expressed in percentage points. The estimates are obtained from an augmented version of specification (2) that includes neutral countries and bilateral indicators for trade between neutral countries and each bloc. The tariff equivalent is obtained by converting the estimated coefficients into ad valorem trade costs using a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t/5) - 1]$. The shaded areas represent 95% confidence intervals, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2, and the vertical dashed line marks the fall of the Berlin Wall.

tion of trade relationships beyond the formal bloc structure itself. Even countries that remained institutionally outside the Cold War alliances exhibited systematic patterns of economic alignment and faced trade barriers that evolved with the broader geopolitical environment.

7 Discussion

The analysis shows that the Cold War division is associated with large trade barriers whose magnitude changed markedly over time. It also shows that measuring these changes requires a complete trade matrix, since standard data miss key flows involving socialist economies. This section interprets the estimated barrier and draws lessons for current debates about how political tensions affect trade.

7.1 What the estimated barrier captures

The estimated barrier between Eastern and Western Europe should not be interpreted as a single, conventional, policy instrument, such as a tariff or a quota. Instead, it is a structural measure of the extra cost of trading across the Cold War divide relative to otherwise comparable trade relationships.

This cost reflects several underlying mechanisms. Some operated through formal barriers, including export controls, licensing requirements, limits on payments, and restrictions on currency convertibility. Others worked through less visible channels: political distrust, weak commercial networks, limited information about foreign buyers and suppliers, and the institutional differences between planned and market economies. The estimate does not separate these channels. It measures their combined effect on trade between Eastern and Western Europe.

This interpretation is useful because many forms of geopolitical fragmentation do not appear as tariffs. During the Cold War, barriers to trade changed with the political relationship between the blocs. They declined during periods of easing tensions, rose during renewed confrontation, and fell sharply near the end of the Cold War. The estimated wedge therefore captures a broad form of trade cost that moved over time with geopolitical events.

A simple quantitative exercise reported in Appendix E suggests that the estimated barriers also have economically meaningful welfare implications. We construct counterfactual simulations using the hat algebra methods popularized by Dekle et al. (2008) and Caliendo and Parro (2015), which require the knowledge of only a limited number of elasticities. Rather than relying solely on a conventional trade elasticity, the exercise allows substitution patterns in the Eastern bloc to differ from those in market economies in order to reflect the institutional organization of socialist

trade. The results imply sizable welfare gains during periods of easing tensions, especially for Eastern European economies, and larger losses for the East when fragmentation increased. Allowing for lower substitution elasticities within the socialist trade system strengthens this conclusion: when economies have fewer close substitutes, disruptions to trade relationships are more costly.

7.2 Implications for fragmentation today

Our evidence from the Cold War period suggests three implications for current debates about trade and political tensions. The first is that fragmentation need not occur through a single policy change. Trade costs can rise through a combination of formal restrictions, limits on technology transfer, payment frictions, weaker commercial links, and greater uncertainty about future political relations. These forces may reduce trade even when tariffs remain unchanged.

Second, fragmentation can evolve gradually. In the Cold War, trade barriers between Eastern and Western Europe did not remain fixed after the division of the continent. They moved over time with the political relationship between the blocs. This matters for the present because changes in trade policy and political relations may affect trade networks slowly, through investment decisions, supplier relationships, and the willingness of firms to enter or maintain foreign markets.

Third, fragmentation changes the direction of integration. During the Cold War, European trade became more integrated within blocs while barriers remained high across the East–West divide. Similar forces can operate today if firms and governments respond to political tensions by shifting trade toward countries with more stable political relationships. The result need not be a uniform decline in trade. It may instead be a reallocation of trade across partners, with some relationships weakening and others deepening.

The historical comparison should be used with care. Today's economy differs from Cold War Europe in important ways: production is more fragmented across countries, supply chains are more complex, services and digital flows are more important, and multinational firms play a larger role. These differences may change the speed and size of adjustment. The Cold War evidence should therefore not be interpreted as a direct guide to current developments. It provides a benchmark for how geopolitical separation can become a measurable trade cost and how that cost can vary over time.

8 Conclusion

This paper estimates the evolution of trade barriers between Eastern and Western Europe during the Cold War within a unified empirical framework. Using newly constructed data on international and domestic trade, we recover a large and time-varying barrier across the East–West divide. The barrier is high in the early Cold War, declines during the period of détente, rises again with the renewed confrontation of the late 1970s and early 1980s, and falls sharply in the final phase of the Cold War. These movements closely track major geopolitical developments, showing that geopolitical fragmentation was not a fixed feature of the period but an evolving constraint on trade.

Accurate measurement is central to this result. Standard international datasets provide incomplete coverage of trade involving Eastern European economies. As a result, they understate the level of trade barriers and attenuate their changes over time. By reconstructing the missing flows, we obtain a structurally interpretable measure of trade barriers. The resulting series reveals substantial variation linked to geopolitical events, a pattern largely missed by standard sources.

The Cold War is a specific historical episode, but it points to a broader lesson: a geopolitical divide can become a large trade cost, and this cost need not be stable over time.

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A Additional Historical Background

A.1 Trade in the socialist bloc

This subsection provides a brief overview of how trade was conducted by socialist economies during the Cold War. A defining feature of the system is the sharp distinction between trade within the socialist bloc (East–East) and trade with Western countries (East–West), both in terms of institutional arrangements and valuation.

A.1.1 East–East trade within the CMEA

Trade among socialist countries was organized within the framework of the Council for Mutual Economic Assistance (CMEA). Rather than relying on decentralized market transactions, trade was planned and coordinated through bilateral agreements specifying quantities and broad terms of exchange. Monetary flows played a limited role, and trade was typically balanced over time.

From the 1960s onward, this system was formalized through multilateral clearing. In 1964, the transferable ruble (TR) was introduced as a non-convertible unit of account for intra-CMEA trade, and remained in use until 1990 (Marrese and Vanous, 1983). Trade flows were recorded in transferable rubles and settled through the Moscow-based International Bank for Economic Cooperation (IBEC), which cleared bilateral balances across member countries.

Prices in East–East trade were not determined at contemporaneous market levels. Instead, administered pricing rules were applied, most notably the "Bucharest price principle," adopted in 1958. Under this system, prices for a given planning period were based on averages of past world market prices; from 1975 onward, a rolling five-year moving average was used. As a result, trade values reflected lagged world prices rather than current market conditions.

In addition, prices were used as an instrument of redistribution within the bloc. Systematic deviations from world market prices—such as relatively low prices for raw materials and relatively high prices for manufactures—have been interpreted as mechanisms for implicit transfers across countries (Marrese and Vanous, 1983). Trade within the CMEA system therefore combined planned allocation, administered pricing, and redistribution.

A.1.2 East–West trade

Trade between socialist economies and Western countries operated under a different regime. Transactions were conducted in convertible currencies (hard currency), such as U.S. dollars,

and settled through standard international payment systems. Prices were generally tied to contemporaneous world market conditions, especially for commodities traded on international markets (Kornai, 1992).

While East–West trade was typically mediated by state trading organizations and subject to political and institutional constraints—such as export controls, bilateral agreements, and credit arrangements—it nevertheless relied on market-based valuation and commercial contracts (Holzman, 1974). Beginning in the mid-1970s, some CMEA countries expanded hard-currency trade among themselves, partially blurring the boundary between the two regimes.

A.1.3 Implications for measurement

These institutional differences have direct implications for the measurement of trade flows. Transactions were recorded in national accounts after conversion into local currency units. Depending on the partner country, trade was settled in convertible currencies (with Western countries), in bilateral clearing currencies (with certain partners), or in transferable rubles (within the CMEA system) (Tarr, 1991).

Planned economies operated systems of multiple exchange rates, with different conversion coefficients applied to different types of transactions (Kornai, 1992). As a result, reported trade values embed country-specific conversion rules. In particular, the conversion of transferable rubles into local currency—and subsequently into U.S. dollars—implies that the same underlying transaction can be assigned different values across reporting countries (Tarr, 1991).

The coexistence of these regimes implies that trade flows involving socialist economies are not uniformly measured. East–East trade reflects a distinct accounting and pricing system based on transferable rubles and administered prices, while East–West trade is much closer to standard market-based trade statistics. This distinction motivates the robustness analysis in Appendix D.3, where we exclude East–East trade flows to assess the sensitivity of our results to the valuation regime specific to intra-CMEA trade.

A.2 Cold War Chronology

Table A.1: Major Geopolitical Phases and Key Events of the Cold War

Event	Date	Notes
1. Bloc Formation and Early Escalation		
Berlin Blockade begins	Jun 1948	First major East–West confrontation in Europe
Berlin Blockade ends	May 1949	Leads to formal division of Germany
NATO established	Apr 1949	Western military alliance
CMEA established	Jan 1949	Eastern economic bloc
Soviet atomic bomb test	Aug 1949	Ends US nuclear monopoly
Korean War begins	Jun 1950	Globalizes Cold War conflict
Stalin dies	Mar 1953	Leadership change in USSR
Korean War armistice	Jul 1953	End of active hostilities
2. Khrushchev Thaw and Renewed Confrontation		
Geneva Summit	Jul 1955	First post-Stalin East–West summit
Austrian State Treaty	May 1955	Austria becomes neutral
Hungarian Uprising	Oct 1956	Soviet intervention signals limits of liberalization
Sputnik launch	Oct 1957	Intensifies technological rivalry
Cuban Revolution	Jan 1959	Expansion of Soviet influence
Berlin Wall constructed	Aug 1961	Physical division of Europe
Cuban Missile Crisis	Oct 1962	Peak nuclear confrontation
3. Détente and Managed Coexistence		
Prague Spring*	Jan 1968	Reform movement in Czechoslovakia
Invasion of Czechoslovakia*	Aug 1968	Reversal of reforms
Brandt takes office	Oct 1969	Start of *Ostpolitik*
SALT I agreement	May 1972	First major arms control agreement
Basic Treaty (FRG–GDR)	Dec 1972	Mutual recognition of East and West Germany
Helsinki Accords	Aug 1975	Framework for East–West cooperation
Vietnam War ends	Apr 1975	Major proxy conflict concludes
USSR invades Afghanistan	Dec 1979	Breakdown of cooperation
4. Renewed Escalation		
Reagan takes office*	Jan 1981	Shift to more confrontational US stance
Strategic Defense Initiative announced	Mar 1983	Escalation of arms race
Gorbachev becomes General Secretary	Mar 1985	Start of reforms (*Perestroika*, *Glasnost*)
5. Late-Cold-War Thaw		
Geneva Summit	Nov 1985	Beginning of renewed engagement
Reykjavik Summit	Oct 1986	Deep arms reduction discussions
INF Treaty signed	Dec 1987	First elimination of nuclear weapons class
6. Collapse of the Eastern Bloc and Transition		
Fall of the Berlin Wall	Nov 1989	Collapse of division in Europe
German reunification	Oct 1990	Formal reunification of Germany
COMECON dissolved	Jun 1991	End of Eastern economic bloc
Dissolution of the USSR	Dec 1991	

Notes: The table summarizes major geopolitical phases of the Cold War and selected key events. Phase boundaries are approximate and reflect sustained changes in the geopolitical environment rather than sharp breaks. Events marked with an asterisk are not interpreted as part of the phase, but included for completeness.

A.3 European bloc definitions

Table A.2: Eastern bloc membership

Country	CMEA/COMECON	Warsaw Pact
Soviet Union	1949 (founder)	1955 (founder)
Poland	1949 (founder)	1955 (founder)
Czechoslovakia	1949 (founder)	1955 (founder)
Hungary	1949 (founder)	1955 (founder)
Romania	1949 (founder)	1955 (founder)
Bulgaria	1949 (founder)	1955 (founder)
East Germany	1950	1956

Notes: The Council for Mutual Economic Assistance (CMEA/COMECON) was established in 1949, and the Warsaw Pact in 1955. The table reports entry years into each institution.

Table A.3: Western bloc membership

Country	Marshall Plan/OEEC	NATO
West Germany	1949	1955
France	1948 (founder)	1949 (founder)
Italy	1948 (founder)	1949 (founder)
Belgium	1948 (founder)	1949 (founder)
Netherlands	1948 (founder)	1949 (founder)
Luxembourg	1948 (founder)	1949 (founder)
United Kingdom	1948 (founder)	1949 (founder)
Denmark	1948 (founder)	1949 (founder)
Norway	1948 (founder)	1949 (founder)
Portugal	1948 (founder)	1949 (founder)
Greece	1948 (founder)	1952

Notes: The Organisation for European Economic Co-operation (OEEC) was created in 1948 to administer the Marshall Plan. NATO was founded in 1949. The table reports entry years into each institution.

Table A.4: Excluded countries and lack of alignment

Country	OEEC	NATO	CMEA	Warsaw Pact	Reason
Spain	No	No*	No	No	Not part of Marshall Plan/OEEC
Austria	Yes	No	No	No	Neutral
Finland	No	No	No	No	Non-aligned
Yugoslavia	No	No	No	No	Non-aligned
Sweden	Yes	No	No	No	Neutral
Switzerland	Yes	No	No	No	Neutral
Ireland	Yes	No	No	No	Neutral
Iceland	Yes	Yes	No	No	Geographically isolated

Notes: The table reports membership status in the relevant institutions. A country is included in the baseline sample if it is clearly aligned with one of the two blocs through participation in the corresponding economic and/or military institutions. Countries are excluded if they lack such alignment (e.g., neutral or non-aligned countries) or, in the case of Iceland, because their geographic isolation provides little identifying variation. *Spain joined NATO in 1982.

B Data Construction Details

B.1 Coverage in the DOTS database

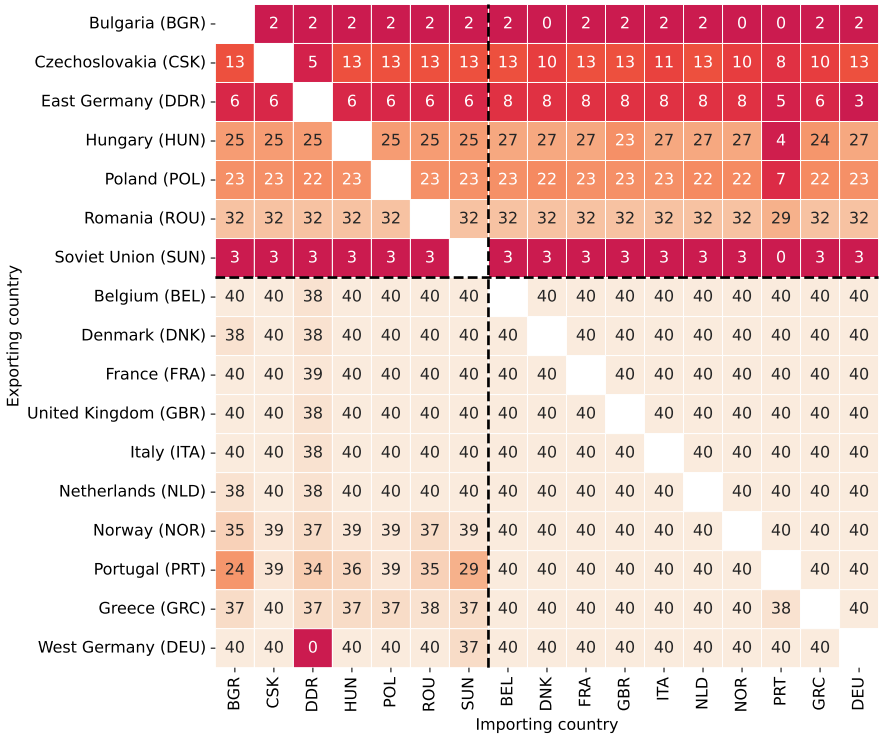


Figure B.1: Coverage of bilateral trade flows in DOTS (1950–1989): exporter-reported flows

Notes: Each cell reports the number of years with non-missing bilateral trade flows for a directed country pair in the DOTS database. Exporting countries are shown in rows and importing countries in columns. The matrix includes only flows reported by the exporting country. A dashed line separates Eastern and Western bloc countries.

Figures B.1 and B.2 provide a disaggregated view of data coverage in the DOTS database by reporting source. A key feature of the data is the limited number of observations directly reported by Eastern bloc countries. In Figure B.1, which focuses on exporter-reported flows, rows corresponding to Eastern economies contain relatively few non-missing observations, indicating that these countries rarely report export data in the DOTS system. A similar pattern emerges in Figure B.2: columns corresponding to Eastern bloc importers exhibit sparse coverage, reflecting limited direct reporting of imports.

As a result, a large share of the observed trade flows involving Eastern bloc countries is derived from mirror statistics reported by their trading partners. This reliance on partner-reported data is evident from the contrast between the two figures: flows involving Eastern economies

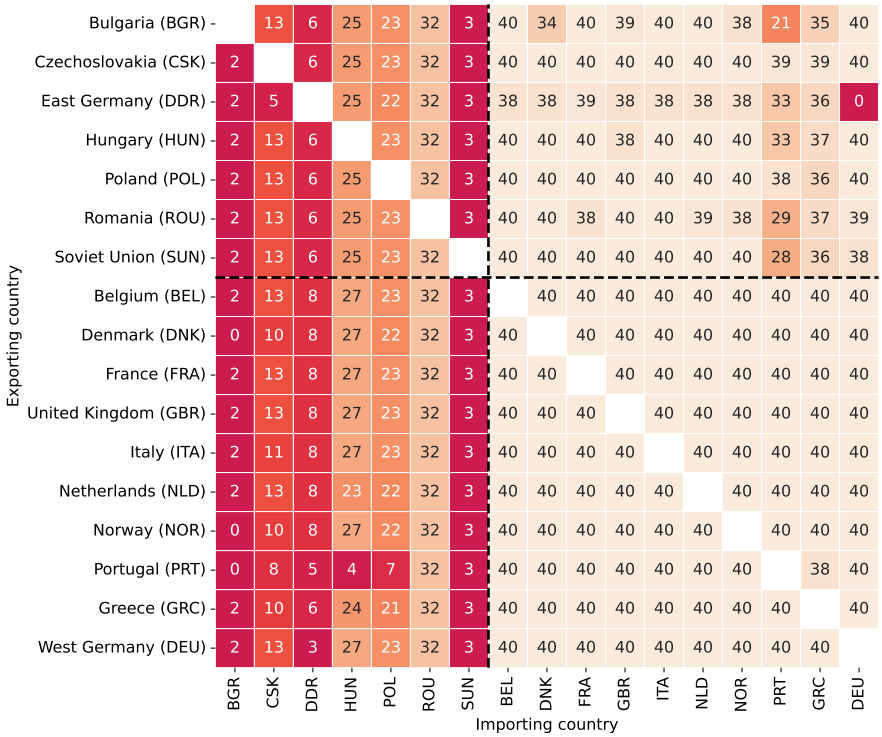


Figure B.2: Coverage of bilateral trade flows in DOTS (1950–1989): importer-reported flows

Notes: Each cell reports the number of years with non-missing bilateral trade flows for a directed country pair in the DOTS database. Exporting countries are shown in rows and importing countries in columns. The matrix includes only flows reported by the importing country. A dashed line separates Eastern and Western bloc countries.

are more frequently observed when reported by Western partners than when reported by the Eastern countries themselves. Consequently, the measurement of trade involving the Eastern bloc depends disproportionately on mirror flows, while direct reporting remains concentrated among Western countries.

B.2 Primary sources

Overview. We complement the DOTS data with bilateral trade flows collected from primary sources for Eastern European countries. These sources provide detailed information on trade with both Eastern and Western partners, often from the perspective of the reporting country. For several countries, the data are obtained from digitized statistical publications, while for others we rely on physical copies of statistical yearbooks consulted in libraries. The data are transformed into a common format and valuation following the DOTS methodology. Despite differences in language, units, and reporting conventions, these sources share a common structure: they report bilateral trade flows by partner country on a consistent annual basis. This makes them well suited for harmonization within a common framework.

Bulgaria. We obtain bilateral trade flows for Bulgaria from the country's statistical yearbooks, published and digitized by the National Statistical Institute.[3] The yearbooks report imports and exports by partner country over the period 1956–1991. Trade flows are reported in lev. Both exports and imports are recorded on a free on board (f.o.b.) basis.

Czechoslovakia. We obtain bilateral trade flows for Czechoslovakia from the country's statistical yearbooks, consulted in physical archives. The yearbooks report imports and exports by partner country over the period 1955–1988. Trade flows are reported in koruna. Exports are recorded on a f.o.b. basis and imports on a c.i.f. basis.

East Germany. We obtain bilateral trade flows for East Germany from the country's statistical yearbooks (*Statistisches Jahrbuch der Deutschen Demokratischen Republik*), which were previously available through the DigiZeitschriften service.[4]

International trade flows are reported in Valutamark, a unit of account used for foreign trade. In early years, trade flows are reported in transferable rubles, but later revisions express these values in Valutamark; we use the revised series. The yearbooks report imports and exports by partner country over the period 1953–1974. For the period 1975–1988, they report only total trade (the sum of exports and imports) by partner country. Both exports and imports are recorded on a free on board (f.o.b.) basis.

[3] https://www.nsi.bg/en/publications. Archived at https://web.archive.org/web/20240610084012/https://www.nsi.bg/en/publications.

[4] https://www.digizeitschriften.de/search?q=514402644. Archived at https://web.archive.org/web/20231006162901/https://www.digizeitschriften.de/search?q=514402644.

Hungary. We obtain bilateral trade flows for Hungary from the country's statistical yearbooks, consulted in physical archives. The yearbooks report imports and exports by partner country for the years 1950, 1955, and over the period 1958–1989. Trade flows are reported in "foreign exchange forints" until 1975 and in forints thereafter. Exports are recorded on a f.o.b. basis and imports on a c.i.f. basis.

Poland. We obtain bilateral trade flows for Poland from the country's statistical yearbooks, consulted in physical archives. The yearbooks report imports and exports by partner country for the years 1950, 1955, and over the period 1960–1988. Trade flows are reported in zloty. Both exports and imports are recorded on a f.o.b. basis until 1980. Starting in 1981, exports are recorded on a f.o.b. basis and imports on a c.i.f. basis.

Soviet Union. We obtain bilateral trade flows for the Soviet Union from the annual foreign trade statistical reviews, which have been digitized and are available online.[5] The statistical reviews report imports and exports by partner country over the period 1950–1988. Trade flows are reported in rubles. Both exports and imports are recorded on a free on board (f.o.b.) basis.

Intra-German trade ("Innerdeutscher Handel"). In addition to East German sources, we use data from the German Statistical Office on trade between East and West Germany. These data were reconstructed after reunification using information from both German states and are available through the GESIS data archive.[6]

The dataset reports bilateral trade flows in Deutsche Mark. Exports are recorded on a f.o.b. basis and imports on a c.i.f. basis. We use the series from Table A.01, which provides trade flows including Berlin for 1952–1989 and excluding Berlin for 1948–1961.

CIA data. We complement the country sources with bilateral trade data from a statistical compilation produced by the Central Intelligence Agency covering the period 1960–1970.[7]

These data are reported directly in current U.S. dollars and are constructed from a combination of official publications, partner-country statistics, and Western estimates. As such, they already reflect the use of official exchange rates consistent with the parity-based system underlying the DOTS methodology during this period.

[5] https://istmat.org/taxonomy/term/344. Archived at https://web.archive.org/web/20230624083918/https://istmat.org/taxonomy/term/344.

[6] https://doi.org/10.4232/1.10259.

[7] Foreign Trade of the East European Communist Countries, 1960–70: A Statistical Summary, declassified CIA Research Aid. Approved for release on 2001/04/12.

Because the CIA compilation is already expressed in U.S. dollars and follows conventions broadly comparable to those used in DOTS, no currency conversion is required. We only apply c.i.f./f.o.b. adjustments where necessary to ensure consistency in the valuation of imports. As a validation check, we compare the CIA series with overlapping observations in DOTS and find that they are closely aligned.

B.3 DOTS harmonization: processing of primary-source trade data

This subsection describes how the bilateral trade data collected from primary sources are transformed into a format comparable to the IMF's Direction of Trade Statistics (DOTS). The objective is to apply, as closely as possible, the same conventions that DOTS would have applied had the underlying observations been directly incorporated into that system.

General procedure. For each source country, we begin from the bilateral trade values reported in the original statistical publications. These values are reshaped into exporter-importer-year format and converted into a common currency using annual exchange rates. We then apply the DOTS convention for the valuation of imports, converting f.o.b. import values into their c.i.f. equivalent whenever the original source reports imports on an f.o.b. basis. The resulting series are expressed in current U.S. dollars and stored in a common format across countries.

Exchange rates. Currency conversion is based on annual external trade conversion factors published separately for exports and imports by the United Nations in its Statistical Yearbook. We use country-specific exchange rates for the currencies appearing in the historical sources, including the valuation of the East German Valutamark, the Bulgarian lev, the Czechoslovak koruna, the Hungarian forint, the Polish zloty, and the Soviet ruble.

For the Soviet Union, we adjust the pre-1961 exchange rate to account for the 1961 currency reform and obtain a consistent series for conversion into U.S. dollars. For Poland, we use the exchange rate reported by the IMF's International Financial Statistics for the year 1981 to avoid a discrepancy between the conversion factors and the way in which trade flows are reported by Polish sources for that particular year.

Valuation and c.i.f./f.o.b. adjustment. Following the DOTS methodology, import flows are expressed on a c.i.f. basis. When the historical source reports imports on an f.o.b. basis, we apply the 10 percent adjustment prescribed in the DOTS methodology for the relevant period. This adjustment is made for Bulgaria, East Germany, and the Soviet Union in all years, and

for Polish imports before 1980. No adjustment is applied when imports are already reported on a c.i.f. basis, as is the case for Hungary, Czechoslovakia, and Polish imports from 1980 onward.

Units of account. The country sources differ in their original units of account and scale. Some report values in millions of local currency units, while others switch units over time. Before conversion to U.S. dollars, all values are transformed into the underlying currency unit corresponding to the exchange rate used.

Country-specific harmonization. The historical sources require a number of source-specific adjustments before they can be merged into a common bilateral dataset.

For Bulgaria, trade with West Germany is split across separate partners (West Berlin and the remainder of West Germany) in the original source; these are aggregated into a single West Germany series before conversion. For Poland, the exchange-rate series is adjusted for the timing with which the official trade statistics recognized the late-1970s devaluation.

For East Germany, bilateral trade flows are reported separately for exports and imports only until 1974. After that date, the published statistics report only the sum of exports and imports at the partner level. As a result, directional bilateral flows for East Germany are available only up to 1974. We incorporate the aggregate series reported after 1974 in a subsequent step of the data construction.

CIA data. We complement the country sources with bilateral trade data from declassified CIA reports, covering the period 1960–1970. These data are reported in millions of U.S. dollars and use exchange-rate conventions consistent with those underlying the DOTS methodology in this period, which are based on official par values.

We assess the comparability of these data by comparing them to DOTS observations for country pairs and years where both sources are available. The two sources align closely, which supports the use of the CIA data to fill gaps in the bilateral trade matrix.

Intra-German trade. Trade between East and West Germany requires separate treatment. In addition to the data reported by East Germany, we use the official post-reunification series published by the Statistisches Bundesamt, which reports bilateral trade between the two German states in Deutsche Mark. These data are converted into U.S. dollars using the corresponding exchange-rate series. Because these flows were reported outside the standard framework of international trade statistics, they are handled separately from the other country sources.

Code harmonization. To merge the reconstructed series with the rest of the trade dataset, partner identifiers in the original sources are harmonized to the ISO-style codes used in the gravity database. In particular, we use SUN for the Soviet Union, CSK for Czechoslovakia, DDR for East Germany, and DEU for West Germany.

The harmonized series produced in this step are then combined with DOTS and other sources using a hierarchical procedure described in the next subsection.

B.4 Construction of bilateral trade flows for estimation

This subsection describes how the different data sources are combined to construct a complete panel of bilateral trade flows for use in the gravity estimation. The procedure follows a hierarchical approach, prioritizing observations according to their valuation (c.i.f. versus f.o.b.) and the reporting country (importer versus exporter).

General approach. For each exporter-importer-year observation, we construct a single measure of trade flows expressed on a c.i.f. basis. When multiple sources are available, we select among them using a fixed priority ordering. This ordering reflects two principles. First, we prioritize data reported by the importing country over data reported by the exporting country, in line with standard practice in the literature. Second, we prioritize observations that are already expressed on a c.i.f. basis over those that require adjustment from f.o.b. values.

Priority ordering of sources. Trade flows are assigned according to the following hierarchy. This hierarchy mirrors the logic of DOTS while exploiting the additional information contained in primary sources.

1. **DOTS (importer-reported, c.i.f.).** We begin with trade flows from DOTS reported by the importing country, which are already expressed on a c.i.f. basis. These observations constitute the benchmark whenever available.

2. **Country sources (importer-reported).** When importer-reported DOTS data are missing, we use the harmonized country-source data when the importing country is one of the Eastern bloc economies in our sample (Bulgaria, Czechoslovakia, Hungary, Poland, the Soviet Union, or East Germany).

3. **CIA data (importer-reported).** When both DOTS and country-source data are unavailable, we use bilateral trade data from declassified CIA sources. These data are treated as importer-reported flows. When the CIA data are reported on an f.o.b. basis,

we convert them to c.i.f. values using the standard 10 percent adjustment for the relevant period.

4. **DOTS (exporter-reported, c.i.f.).** If no importer-reported data are available, we use c.i.f. values from DOTS derived from the exporting country.

5. **DOTS (f.o.b., adjusted to c.i.f.).** If only f.o.b. values are available in DOTS, we convert them to c.i.f. values using the DOTS methodology. We apply a 10 percent adjustment for years before 2000 and a 6 percent adjustment thereafter, following the change in the DOTS convention.

6. **Country sources (exporter-reported).** As a final step, when no other information is available, we use the harmonized country-source data when the exporting country is one of the Eastern bloc economies.

Intra-German trade. Trade between East and West Germany is treated separately from the general hierarchy. Whenever available, we replace all other sources with the bilateral flows constructed from the German statistical sources ("Innerdeutscher Handel"). These data are considered the most reliable measure of trade between the two German states, as they are based on reconciled information from both reporting systems. They are used regardless of the availability of alternative observations in DOTS or other sources.

Output. This procedure yields a complete panel of bilateral trade flows expressed on a c.i.f. basis, together with an indicator of the underlying data source. The resulting dataset maximizes coverage while maintaining a consistent valuation and prioritizing the most reliable observations available for each country pair and year.

B.5 Domestic trade: construction details

This subsection describes the construction of domestic trade flows used in the main text. We closely follow Head and Mayer (2021) and document the adjustments required for our historical sample.

Definition. Domestic trade is defined as

$$X_{iit} = Q_{it} - X_{it},$$

where Q_{it} denotes total production of tradable goods in country i and year t, and X_{it} denotes total exports. Following Head and Mayer (2021), goods production is expressed as a share of GDP:

$$Q_{it} = \kappa_{it} \cdot GDP_{it},$$

where κ_{it} is the ratio of goods production (agriculture, mining, and manufacturing) to GDP. Combining the two expressions yields

$$X_{iit} = \kappa_{it} \cdot GDP_{it} - X_{it}.$$

Baseline construction. The series for κ_{it} is constructed from sectoral data on output and value added. When direct measures of output are unavailable, output-to-GDP ratios are interpolated across time and extrapolated using available information on sectoral composition. Aggregate exports X_{it} are taken directly from the Head and Mayer (2021) dataset, which integrates Comtrade data when available and complements it with IMF DOTS otherwise.

We implement this procedure using the replication files of Head and Mayer (2021). Our Python translation reproduces their original dataset exactly.

Adjustments for the historical sample. We make a limited number of adjustments to adapt the baseline construction to our setting.

GDP series. We combine the GDP series used in Head and Mayer (2021) with historical GDP data from Barbieri (1996). For defunct countries (the Soviet Union, Czechoslovakia, Yugoslavia, and East Germany), we rely exclusively on the Barbieri data to avoid discontinuities associated with changes in sources.

For the remaining countries, we extend GDP series backward using growth rates from the Barbieri data. Specifically, when GDP is missing in early years, we rescale the Barbieri series to match the first available observation in the baseline data and use its implied growth rates to backfill the series.

Trade data for East Germany. Aggregate trade data for East Germany are mismeasured in both Comtrade and DOTS. We therefore replace total exports for East Germany with values constructed from primary sources, as described in the main text. These corrected series are used in the computation of domestic trade.

Other adjustments. First, we retain historical entities (e.g., Czechoslovakia, the Soviet Union, Yugoslavia) rather than mapping them to modern successor states.

Second, for several Eastern European countries, the series for κ_{it} does not extend to the beginning of our sample. We extend these series back to 1950 by extrapolating the earliest available observations.

Finally, for East Germany, no initial observation of κ_{it} is available from which to extrapolate. We therefore impute κ_{it} using the contemporaneous average across other Eastern bloc economies (Bulgaria, Czechoslovakia, Hungary, Poland, Romania, and the Soviet Union):

$$\kappa_{DDR,t} = \frac{1}{6} \sum_{j \in \mathcal{E}} \kappa_{jt},$$

where $\mathcal{E}$ denotes the set of Eastern bloc countries excluding East Germany.

C Identification in the Two-Bloc Gravity Model

This appendix formally characterizes the identifying variation in the bloc-restricted gravity model described in the main text. To focus on the core logic, we suppress the time dimension and bilateral controls and consider a static setting with exporter and importer fixed effects.

Let each country i belong either to the East or the West. Define

$$E_i \equiv \mathbf{1}\{i \in East\}, \qquad W_i \equiv 1 - E_i.$$

For any ordered pair (i, j), define the bloc-pair indicators

$$EE_{ij} = E_i E_j, \quad EW_{ij} = E_i W_j, \quad WE_{ij} = W_i E_j, \quad WW_{ij} = W_i W_j.$$

These satisfy

$$EE_{ij} + EW_{ij} + WE_{ij} + WW_{ij} = 1.$$

Define the cross-bloc indicator

$$CB_{ij} \equiv EW_{ij} + WE_{ij}.$$

We consider linear predictors of the form

$$\eta_{ij} = \phi_i + \psi_j + \text{bloc-pair terms},$$

where ϕ_i and ψ_j denote exporter and importer fixed effects.

C.1 Identification with and without domestic trade

Proposition 1 (Identification of bloc-pair effects)**.** *Consider the bloc-restricted gravity model with exporter and importer fixed effects.*

(i) In the sample of international observations only ($i \neq j$), the bloc-pair indicators collapse to a one-dimensional space once exporter and importer fixed effects are included. Only the cross-bloc indicator CB_{ij} is identified, capturing cross-bloc trade relative to within-bloc trade.

(ii) In the sample that includes both international and domestic observations ($i = j$), with domestic trade as the omitted category, the indicators $(CB_{ij}, EE_{ij}, WW_{ij})$ are linearly independent conditional on exporter and importer fixed effects. Hence cross-bloc, East–East, and West–West trade are separately identified relative to domestic trade.

(iii) In either case, the directional asymmetry between East-to-West and West-to-East trade is not identified. Only the symmetric component $CB_{ij} = EW_{ij} + WE_{ij}$ is identified.

Intuition. The key insight is that exporter and importer fixed effects absorb all country-specific variation, leaving only genuinely bilateral patterns identified. In a two-bloc setting with international trade only, the distinction between East and West on the exporter and importer sides can be fully captured by these fixed effects. As a result, the four bloc-pair categories (East–East, East–West, West–East, West–West) collapse to a single dimension: whether trade crosses the bloc boundary or not. Including domestic trade introduces an additional benchmark that cannot be written as a combination of exporter and importer effects, because domestic observations lie on the diagonal of the trade matrix. This breaks the previous collinearity and allows the model to distinguish between cross-bloc trade, East–East trade, and West–West trade, each relative to domestic trade. However, directional differences between East-to-West and West-to-East trade remain unidentifiable, since they can be expressed entirely in terms of exporter and importer characteristics.

Proof. **Part (i): International trade only.**

Restrict attention to $i \neq j$. Consider the saturated specification

$$\eta_{ij} = \phi_i + \psi_j + \beta_{EE} EE_{ij} + \beta_{EW} EW_{ij} + \beta_{WE} WE_{ij} + \beta_{WW} WW_{ij}.$$

Using the identities

$$E_i = EE_{ij} + EW_{ij}, \qquad E_j = EE_{ij} + WE_{ij},$$

we can write

$$EW_{ij} = E_i - EE_{ij}, \qquad WE_{ij} = E_j - EE_{ij},$$

and

$$WW_{ij} = 1 - E_i - E_j + EE_{ij}.$$

Substituting into η_{ij} yields

$$\eta_{ij} = [\phi_i + (\beta_{EW} - \beta_{WW})E_i] + [\psi_j + (\beta_{WE} - \beta_{WW})E_j] + (\beta_{EE} - \beta_{EW} - \beta_{WE} + \beta_{WW})EE_{ij} + \beta_{WW}.$$

The terms involving E_i and E_j are absorbed by exporter and importer fixed effects, and the constant is irrelevant. Therefore only a single linear combination of bloc-pair indicators remains identified. Since

$$CB_{ij} = 1 - EE_{ij} - WW_{ij},$$

and EE_{ij} and WW_{ij} cannot be separately identified once fixed effects are included, the remaining variation can equivalently be represented by CB_{ij}. This establishes that only one bilateral contrast survives.

Part (ii): Including domestic trade.

Now include domestic observations $i = j$, and define

$$EE_{ij} = \mathbf{1}\{i \in East,\ j \in East,\ i \neq j\}, \quad WW_{ij} = \mathbf{1}\{i \in West,\ j \in West,\ i \neq j\},$$

with domestic trade as the omitted category, so that

$$CB_{ii} = EE_{ii} = WW_{ii} = 0.$$

Let $D_{ij} \equiv \mathbf{1}\{i = j\}$ denote the domestic indicator. Then

$$D_{ij} + CB_{ij} + EE_{ij} + WW_{ij} = 1.$$

Suppose that a linear combination of $(CB_{ij}, EE_{ij}, WW_{ij})$ can be absorbed by exporter and importer fixed effects:

$$\lambda_{CB} CB_{ij} + \lambda_{EE} EE_{ij} + \lambda_{WW} WW_{ij} = a_i + b_j.$$

We show that this implies $\lambda_{CB} = \lambda_{EE} = \lambda_{WW} = 0$.

Take distinct Eastern countries $e \neq e'$ and Western countries $w \neq w'$. Using domestic observations,

$$0 = a_e + b_e, \qquad 0 = a_w + b_w.$$

Using cross-bloc observations,

$$\lambda_{CB} = a_e + b_w = a_w + b_e.$$

Substituting $b_e = -a_e$ and $b_w = -a_w$ gives

$$\lambda_{CB} = a_e - a_w = a_w - a_e,$$

which implies $a_e = a_w$ and hence $\lambda_{CB} = 0$.

Using East–East observations,

$$\lambda_{EE} = a_e + b_{e'} = a_e - a_{e'},$$

and by symmetry also $\lambda_{EE} = a_{e'} - a_e$, implying $\lambda_{EE} = 0$. The same argument applied to West–West observations yields $\lambda_{WW} = 0$.

Therefore, no nontrivial linear combination of $(CB_{ij}, EE_{ij}, WW_{ij})$ lies in the span of exporter and importer fixed effects, so all three coefficients are identified.

Part (iii): Directional asymmetry.

Consider the difference between East-to-West and West-to-East indicators:

$$EW_{ij} - WE_{ij} = E_i W_j - W_i E_j.$$

Using $W_i = 1 - E_i$ and $W_j = 1 - E_j$, this simplifies to

$$EW_{ij} - WE_{ij} = E_i - E_j.$$

Since E_i is exporter-specific and E_j is importer-specific, the difference $EW_{ij} - WE_{ij}$ is absorbed by exporter and importer fixed effects. Therefore, only the symmetric component

$$CB_{ij} = EW_{ij} + WE_{ij}$$

is identified.

□

Implications for the empirical specifications. Proposition 1 clarifies the interpretation of the coefficients in the main text. In the international-only specification (1), the coefficient γ_t^{CB} is the only bilateral bloc contrast that survives once exporter-year and importer-year fixed effects are included, and it should therefore be interpreted as the cross-bloc trade wedge relative to within-bloc international trade. In the augmented specification (2), the inclusion of domestic trade allows separate identification of γ_t^{CB}, γ_t^{EE}, and γ_t^{WW}, which can be interpreted as border effects relative to domestic trade. In particular, γ_t^{CB} captures the degree of fragmentation between the two blocs, expressed as the trade cost associated with crossing the bloc boundary relative to trading at home.

C.2 Domestic trade as an implicit normalization

To clarify the role of domestic trade in the augmented specification, it is useful to write a more general model that allows for an additional latent domestic component. Consider

$$X_{ijt} = \exp\Big(\gamma_t^{CB} CB_{ij} + \gamma_t^{EE} EE_{ij} + \gamma_t^{WW} WW_{ij} + \phi_{it} + \psi_{jt} + z'_{ij}\beta + h_{it}\,\mathbf{1}\{i = j\}\Big) + \varepsilon_{ijt}, \tag{C.1}$$

where h_{it} is a country-year-specific domestic-trade component that applies only on the diagonal of the trade matrix.

Equation (C.1) nests the baseline domestic-trade specification in the main text. In particular, the empirical specification (2) is obtained by imposing

$$h_{it} = 0 \qquad \text{for all } i, t. \tag{C.2}$$

This restriction does *not* imply that domestic trade is equal across countries or over time in levels. Such differences may still arise through the exporter-year and importer-year fixed effects, ϕ_{it} and ψ_{jt}. Rather, (C.2) imposes a more specific normalization: conditional on exporter-year effects, importer-year effects, and bilateral controls, there is no additional residual domestic-trade component that is specific to country i in year t.

Equivalently, domestic observations are treated as a common omitted bilateral benchmark. For domestic flows ($i = j$), the bloc-pair indicators satisfy

$$CB_{ii} = EE_{ii} = WW_{ii} = 0,$$

so under (2) the latent index for domestic trade is

$$\eta_{iit} = \phi_{it} + \psi_{it} + z'_{ii}\beta.$$

Thus, after partialling out the fixed effects structure, all remaining bilateral differences are measured relative to the same normalized domestic benchmark.

Implications for the level of the cross-bloc coefficient. To see how this affects the interpretation of γ_t^{CB}, consider an Eastern country e and a Western country w. Under the general specification (C.1),

$$\eta_{ew,t} = \phi_{et} + \psi_{wt} + \gamma_t^{CB}, \qquad \eta_{we,t} = \phi_{wt} + \psi_{et} + \gamma_t^{CB},$$

while for the domestic observations,

$$\eta_{ee,t} = \phi_{et} + \psi_{et} + h_{et}, \qquad \eta_{ww,t} = \phi_{wt} + \psi_{wt} + h_{wt}.$$

Taking the symmetric cross-bloc contrast relative to domestic trade gives

$$\frac{1}{2}\Big[(\eta_{ew,t} + \eta_{we,t}) - (\eta_{ee,t} + \eta_{ww,t})\Big] = \gamma_t^{CB} - \frac{1}{2}(h_{et} + h_{wt}). \tag{C.3}$$

Equation (C.3) shows that the cross-bloc coefficient has a border-effect interpretation relative to domestic trade only if the residual domestic component is normalized away. If $h_{it} \neq 0$, the level of γ_t^{CB} conflates the cross-bloc wedge with the average residual ease of domestic trade in the countries used to form the comparison.

Implications for the evolution of the cross-bloc coefficient. The same logic applies over time. Let t and s denote two years. Then the change in the symmetric cross-bloc contrast relative to domestic trade is

$$\Delta\left\{\frac{1}{2}\Big[(\eta_{ew} + \eta_{we}) - (\eta_{ee} + \eta_{ww})\Big]\right\} = \Delta\gamma^{CB} - \frac{1}{2}(\Delta h_e + \Delta h_w), \tag{C.4}$$

where $\Delta h_e \equiv h_{et} - h_{es}$ and $\Delta h_w \equiv h_{wt} - h_{ws}$.

Hence the time path of γ_t^{CB} can be interpreted as the evolution of cross-bloc trade frictions relative to domestic trade only if the residual domestic component does not vary over time in ways not absorbed by the fixed effects. If residual domestic trade becomes easier or harder over time, then changes in γ_t^{CB} reflect both changes in cross-bloc fragmentation and changes in the domestic benchmark.

Interpretation. The augmented specification with domestic trade therefore relies on a stronger normalization than the international-only specification. The exporter-year and importer-year fixed effects allow domestic trade to differ across countries and over time through general country-year factors. However, they do not absorb an additional country-year-specific premium that applies only to domestic trade. By imposing (C.2), the model treats such diagonal-specific variation as absent. Under this restriction, γ_t^{CB} is interpretable as the cross-bloc trade wedge relative to domestic trade. If the restriction fails, the level and evolution of γ_t^{CB} also reflect shifts in the residual domestic-trade component.

C.3 Identification after removing East–East trade

This subsection characterizes identification when international trade between Eastern bloc countries is excluded from the sample. We suppress the time dimension and bilateral controls and work in a static setting with exporter and importer fixed effects.

We retain the notation

$$E_i = \mathbf{1}\{i \in East\}, \qquad W_i = 1 - E_i,$$

and define

$$CB_{ij} = \mathbf{1}\{i \in East,\ j \in West\} + \mathbf{1}\{i \in West,\ j \in East\},$$

$$WW_{ij} = \mathbf{1}\{i \in West,\ j \in West,\ i \neq j\}.$$

Throughout this subsection, we restrict the sample by excluding all East–East international observations ($EE_{ij} = 1$ with $i \neq j$). We study identification in the specification

$$\eta_{ij} = \phi_i + \psi_j + \gamma^{CB} CB_{ij} + \gamma^{WW} WW_{ij}.$$

Proposition 2 (Identification after removing East–East trade)**.** *Consider the bloc-restricted gravity model with exporter and importer fixed effects, after excluding all East–East international observations.*

(i) If domestic trade for both Eastern and Western countries is retained, then CB_{ij} and WW_{ij} are linearly independent conditional on exporter and importer fixed effects. Hence γ^{CB} and γ^{WW} are separately identified.

(ii) If domestic trade for Eastern countries is also excluded (while domestic trade for Western countries is retained), then CB_{ij} lies in the span of exporter and importer fixed effects and is therefore not identified. In this case, γ^{CB} cannot be estimated, and CB_{ij} and WW_{ij} cannot be jointly identified.

Intuition. The key distinction is whether the sample contains observations in which an Eastern country appears on both sides of the dyad without crossing the bloc boundary. When domestic trade for Eastern countries is retained, such observations exist (the diagonal of the trade matrix), and they break the link between cross-bloc status and exporter/importer bloc membership. This allows CB_{ij} to capture a genuinely bilateral pattern that is not absorbed by fixed effects.

By contrast, when both East–East international trade and Eastern domestic trade are excluded, every observation involving an Eastern country is necessarily cross-bloc. In that case, the

cross-bloc indicator becomes equivalent to the presence of an Eastern country on either side of the dyad, which is a purely exporter/importer characteristic and is therefore absorbed by the fixed effects.

Proof. **Part (i): Retaining domestic trade for Eastern countries.**

Suppose there exist constants $\lambda_{CB}, \lambda_{WW}$ and functions a_i, b_j such that

$$\lambda_{CB} CB_{ij} + \lambda_{WW} WW_{ij} = a_i + b_j$$

for all observations in the sample.

Let e denote an Eastern country and w, w' denote distinct Western countries. Using domestic observations,

$$0 = a_e + b_e, \qquad 0 = a_w + b_w, \qquad 0 = a_{w'} + b_{w'}.$$

Using cross-bloc observations,

$$\lambda_{CB} = a_e + b_w = a_w + b_e.$$

Substituting $b_e = -a_e$ and $b_w = -a_w$ yields

$$\lambda_{CB} = a_e - a_w = a_w - a_e,$$

which implies $a_e = a_w$ and hence $\lambda_{CB} = 0$.

Using a West–West international observation,

$$\lambda_{WW} = a_w + b_{w'} = a_w - a_{w'},$$

and reversing the pair,

$$\lambda_{WW} = a_{w'} + b_w = a_{w'} - a_w.$$

This implies $a_w = a_{w'}$ and hence $\lambda_{WW} = 0$.

Therefore, the only solution is $\lambda_{CB} = \lambda_{WW} = 0$, establishing that CB_{ij} and WW_{ij} are linearly independent conditional on exporter and importer fixed effects.

Part (ii): Excluding domestic trade for Eastern countries.

Now suppose that domestic observations for Eastern countries are removed, while domestic observations for Western countries are retained.

On the remaining sample, any observation involving an Eastern country must be cross-bloc. Therefore, for all (i, j) in the sample,

$$CB_{ij} = E_i + E_j.$$

Since E_i depends only on the exporter and E_j depends only on the importer, it follows that

$$CB_{ij} = E_i + E_j$$

lies in the span of exporter and importer fixed effects. Hence CB_{ij} is perfectly collinear with the fixed effects and is not identified.

It follows that γ^{CB} cannot be estimated, and CB_{ij} cannot be jointly identified with WW_{ij}.

□

Implications for the empirical analysis. Proposition 2 clarifies the role of Eastern domestic trade in identifying cross-bloc trade effects when East–East trade is excluded. If domestic trade for Eastern countries is retained, the specification continues to identify both the cross-bloc coefficient γ_t^{CB} and the West–West coefficient γ_t^{WW} relative to domestic trade. In contrast, if Eastern domestic trade is also excluded, the cross-bloc indicator is absorbed by exporter-year and importer-year fixed effects, and γ_t^{CB} is no longer identified. This highlights that some within-East observations—either East–East international trade or Eastern domestic trade—are necessary to distinguish bilateral cross-bloc patterns from purely exporter- or importer-specific effects.

D Additional Empirical Results

D.1 Within-bloc trade costs

This appendix subsection reports estimates of within-bloc trade costs for the Eastern and Western blocs. These estimates complement the cross-bloc results discussed in the main text by documenting the evolution of economic integration within each bloc. Trade costs decline substantially over the Cold War period in both cases, consistent with increasing integration. The decline is relatively smooth among Western countries, while estimates for the Eastern bloc exhibit greater variation, reflecting both genuine fluctuations and more limited data availability. In particular, intra-Eastern-bloc trade costs display a pronounced spike following 1989, consistent with a disruption of established trading relationships during the transition period, although measurement issues related to domestic output may also contribute.

In addition to this post-1989 episode, intra-bloc trade costs within the Eastern bloc rise noticeably between the early 1950s and the end of that decade. This period coincides with a phase of transition and consolidation following the death of Joseph Stalin in 1953. The subsequent reorientation of economic policy and the gradual restructuring of production and trade relationships likely generated temporary frictions as countries adjusted to new patterns of coordination. Episodes of political instability within the bloc, most notably the Hungarian Revolution of 1956, may have further disrupted economic activity. Trade costs decline again toward the end of the 1950s, consistent with a gradual stabilization of economic relations within the bloc. As with other episodes, part of the variation may also reflect measurement challenges in the early years of the sample, and we therefore interpret these fluctuations with caution.

A further increase in Eastern intra-bloc trade costs is visible in the early 1980s. This pattern is consistent with the broader economic and political difficulties affecting Eastern Europe during this period. Several countries experienced significant disruptions, most notably Poland following the rise of the Solidarity movement in 1980 and the imposition of martial law in 1981, which affected production and trade flows. More generally, many Eastern European economies faced rising external imbalances, debt pressures, and stagnating growth, leading to adjustments that disrupted established trading relationships within the bloc. These developments likely contributed to the observed increase in trade costs. At the same time, given the challenges of measuring domestic economic activity during periods of stress, part of this pattern may also reflect measurement issues, and we therefore interpret it with caution.

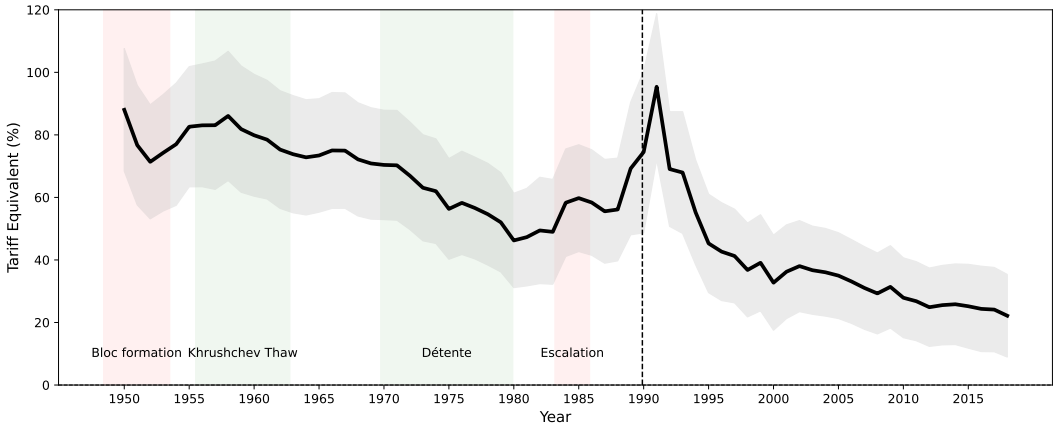


Figure D.1: Within-bloc trade costs: Eastern bloc

Notes: The figure shows the estimated tariff equivalent of trade between countries in the Eastern bloc (East–East), expressed in percentage points. The estimates are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is obtained by converting the estimated coefficient $\hat{\gamma}_t^{EE}$ into an ad valorem trade cost using a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t^{EE}/5) - 1]$. The shaded area represents the 95% confidence interval, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2, and the vertical dashed line marks the fall of the Berlin Wall.

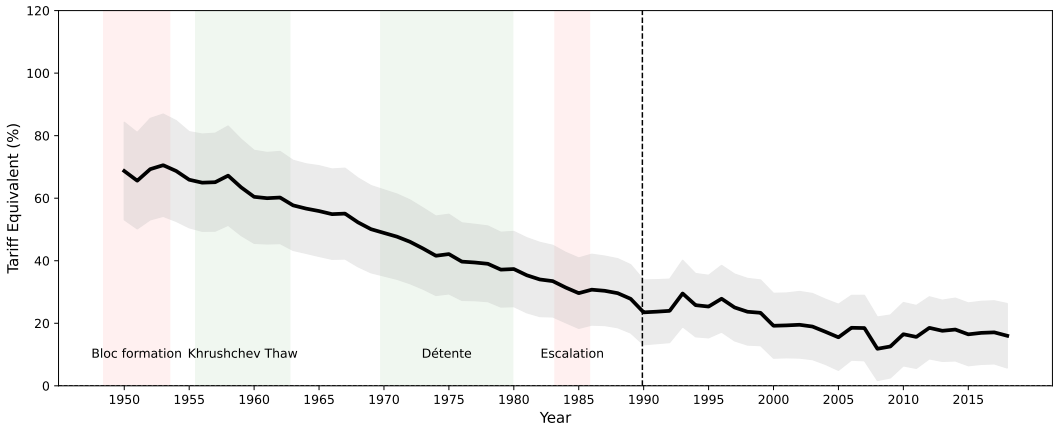


Figure D.2: Within-bloc trade costs: Western bloc

Notes: The figure shows the estimated tariff equivalent of trade between countries in the Western bloc (West–West), expressed in percentage points. The estimates are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is obtained by converting the estimated coefficient $\hat{\gamma}_t^{WW}$ into an ad valorem trade cost using a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t^{WW}/5) - 1]$. The shaded area represents the 95% confidence interval, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2, and the vertical dashed line marks the fall of the Berlin Wall.

D.2 Domestic trade robustness

A key input into the estimation is the construction of domestic trade (self-trade). In the baseline, domestic trade is inferred from national accounts using time-varying goods-production shares (κ_{it}) and total output. While this approach is standard, it relies on auxiliary data that may be measured with error, particularly for centrally planned economies and in earlier decades.

To assess the robustness of our results, we construct four alternative measures of domestic trade. Each is designed to address a distinct potential concern.

First, we consider a *fixed tradables-share specification*, in which the goods-production share of each country (κ_{it}) is held constant up to 1991 at its average value over the 1970s, and allowed to vary freely thereafter. The 1970s are chosen as a relatively stable period that is also used to calibrate the trade matrix in the general equilibrium model used in the discussion in Section 7. For East Germany, we assign the corresponding East-bloc average. This specification isolates the role of time variation in sectoral composition during the Cold War, and guards against the possibility that such variation—especially if mismeasured—induces spurious movements in domestic trade and, in turn, in the estimated border effect.

Second, we construct a *smoothed-production specification*, where output entering the domestic-trade calculation is replaced by a three-year moving average. This addresses concerns about short-run volatility or measurement error in national accounts data, which may be particularly pronounced in historical series and for countries undergoing structural transformation.

Third, we implement a *Cold War GDP backbone*, combining historical GDP data by Barbieri (1996) for all years up to 1991 with the baseline GDP series thereafter. For each country, Barbieri GDP is scaled to match the level of the baseline series at the first overlapping observation, so that only growth rates are imported. This specification is intended to mitigate concerns about inconsistencies in national accounts data across countries and over time, especially in the early Cold War period and for centrally planned economies.

Fourth, we construct an *East-bloc tradables-share specification*, in which country-specific goods-production shares for Eastern bloc economies are replaced by the contemporaneous average across Eastern bloc countries. This specification is based on the completed goods-production-share series used in the baseline construction, and is designed to reduce the influence of country-specific measurement error in Eastern Europe while preserving the common time variation in sectoral composition within the bloc.

In all cases, domestic trade is recomputed using the same accounting identity as in the baseline, replacing only the relevant component (tradables share or output). The gravity equation is then re-estimated without further modification.

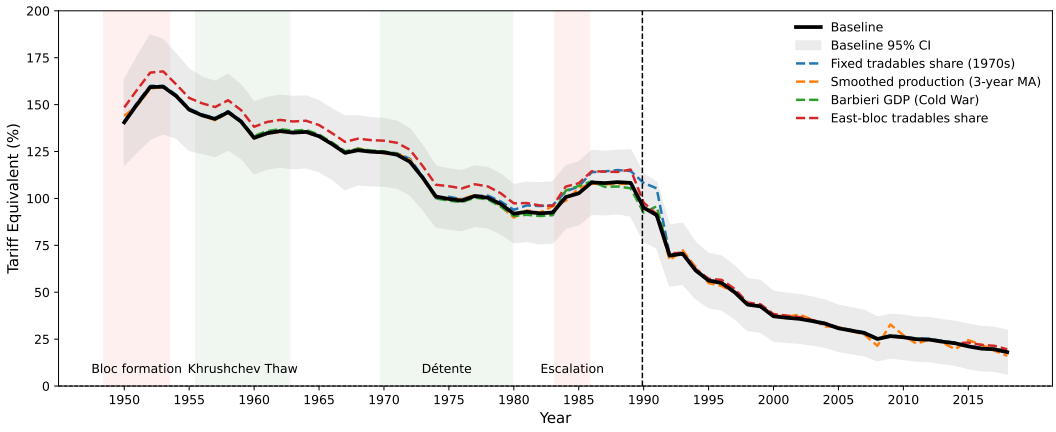


Figure D.3: Domestic trade robustness: cross-bloc barrier

Notes: The figure reports the estimated tariff-equivalent of the cross-bloc trade barrier under alternative constructions of domestic trade. The baseline measure uses time-varying goods-production shares and output. The four alternative specifications are: (i) a fixed tradables share, where goods-production shares are held constant at their 1970s average up to 1991; (ii) smoothed production, where output is replaced by a three-year moving average; (iii) a Cold War GDP backbone, where Barbieri (1996) GDP data are used up to 1991 and scaled to match baseline levels, with the baseline series used thereafter; and (iv) an East-bloc tradables share, where country-specific goods-production shares for Eastern bloc economies are replaced by the contemporaneous East-bloc average. All specifications are estimated using the same gravity equation with exporter-year and importer-year fixed effects. The shaded area represents the 95% confidence interval, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2. A vertical dashed line marks the fall of the Berlin Wall.

The results are highly stable across specifications. Three alternatives are nearly indistinguishable from the baseline over most of the Cold War period, whereas the specification based on the East-bloc tradables share produces estimates that are systematically higher in levels, but closely track the baseline over time. The fixed tradables-share specification deviates from the baseline only briefly during the transition period following the fall of the Berlin Wall. Overall, the medium-run evolution and the timing of turning points are preserved across all variants.

These results indicate that the construction of domestic trade primarily affects the level of the estimated border effect, but not its evolution. The medium-run pattern and the timing of turning points are preserved across all variants, reinforcing the conclusion that the cross-bloc trade barrier evolves in a stable and economically meaningful way over time.

An important intuition for these results comes from the structure of the gravity model with exporter-year and importer-year fixed effects. These fixed effects absorb all variation in total outward and inward trade for each country-year, implying that identification relies on how trade is allocated across partners rather than on aggregate trade volumes. In particular, domestic trade corresponds to the diagonal of the bilateral trade matrix. Alternative constructions of domestic trade therefore reallocate mass between domestic and international flows within a country-year, shifting the relative position of the diagonal entries and, in turn, the level of the estimated border effect.

The dynamics, however, are preserved as long as differences across domestic trade measures are relatively stable over time, so that the relative evolution of domestic versus foreign trade remains similar across specifications. Under this condition, alternative constructions do not materially affect the time pattern of the estimated border effect. This mechanism is consistent with the findings of Campos et al. (2021), who show that different methods of constructing domestic trade yield very similar estimates in structural gravity models, as most of the variation across methods is absorbed by the fixed effects.

D.3 Valuation and pricing distortions in East–East trade

The main comparability concerns in our setting arise from systematic differences between trade within the socialist bloc (East–East) and trade with Western countries (East–West). East–West trade was typically conducted in convertible currencies, priced at contemporaneous world market prices, and governed by standard commercial arrangements. By contrast, East–East trade within the Council for Mutual Economic Assistance (CMEA)—the economic organization of socialist countries—operated under a distinct institutional and accounting regime. We focus on three features of this regime: (i) transferable rubles, (ii) administered pricing rules such as the *Bucharest formula*, and (iii) pricing-based redistribution across countries.

Transferable ruble conversion. East–East trade was denominated in transferable rubles (TR), a non-convertible unit of account used in the CMEA clearing system. Transactions were recorded in national accounts by converting TR flows into local currency units (LCU) using official, country-specific rates unrelated to market exchange rates. As a result, there is no unique mapping from transferable rubles into U.S. dollars.

By contrast, East–West trade was conducted in convertible currencies and valued using standard procedures. Identical physical transactions could therefore be assigned different values depending on whether they were recorded within the CMEA system or in market-based trade statistics. In our data, this problem arises upstream, since reported trade values already embed these conversion choices.

Administered pricing and the Bucharest formula. Prices in East–East trade were generally not set at contemporaneous world market levels. Many transactions—especially for homogeneous goods—were priced using the Bucharest formula, which relied on moving averages of past world prices, while others were determined through bilateral planning arrangements. As a result, East–East trade values reflect lagged and administered prices.

In contrast, East–West trade was priced at current world market prices. When world prices changed, identical flows could therefore be valued differently across the two regimes, generating systematic divergences unrelated to quantities.

Implicit transfers through administered prices. Administered prices in East–East trade often embodied redistribution across CMEA members. Sectoral distortions—such as relatively low prices for raw materials and relatively high prices for manufactures—have been interpreted as mechanisms for implicit transfers within the bloc. These were implemented through deviations from world price benchmarks.

Such systematic redistribution was not a feature of East–West trade. Consequently, East–East trade values reflect not only exchange but also politically determined transfers that vary across countries and over time.

Robustness to intra-CMEA valuation regimes. To address these concerns, we re-estimate the model excluding all East–East trade flows while retaining domestic trade for all countries.

Because transferable-ruble accounting, administered pricing, and implicit transfers were specific to the CMEA system, excluding East–East flows removes this valuation regime altogether. The remaining East–West and West–West flows are conducted in convertible currencies and priced at contemporaneous world prices, providing a more comparable basis for analysis.

Retaining domestic trade for Eastern countries is essential for identification. As shown in Appendix C.3, without Eastern domestic observations the cross-bloc indicator is collinear with exporter-time and importer-time fixed effects. Domestic trade therefore provides the within-East variation needed to identify bilateral cross-bloc effects.

This approach has two caveats. First, multilateral resistance terms are now computed over a reduced trade network that excludes East–East links. While this departs from a full general-equilibrium representation, it may be preferable if East–East flows are not consistently valued. Second, the specification continues to rely on constructed domestic trade for Eastern countries, which may be measured with error.

Figure D.4 compares the estimated cross-bloc trade barrier in the baseline specification to the estimates obtained after excluding all international East–East flows. The two series are remarkably similar throughout the sample period. In particular, the level and time profile of the estimated cross-bloc wedge are largely unaffected by the exclusion of intra-CMEA trade.

This stability indicates that the distinctive accounting and pricing features of East–East trade do not materially influence the estimated fragmentation effect. Instead, the identifying variation for the cross-bloc coefficient appears to arise primarily from comparisons between cross-bloc and market-based trade flows (East–West and West–West), rather than from the internal structure of trade within the socialist bloc.

At the same time, the exclusion of East–East flows leads to a modest increase in standard errors. This suggests that these observations contribute to estimation by providing additional support for exporter-year and importer-year fixed effects in a more complete trade matrix, thereby improving precision without materially affecting the point estimates.

Taken together, these results provide evidence that the baseline estimates are not driven by the valuation and pricing conventions specific to intra-CMEA trade, but instead reflect a broader and robust pattern of cross-bloc fragmentation.

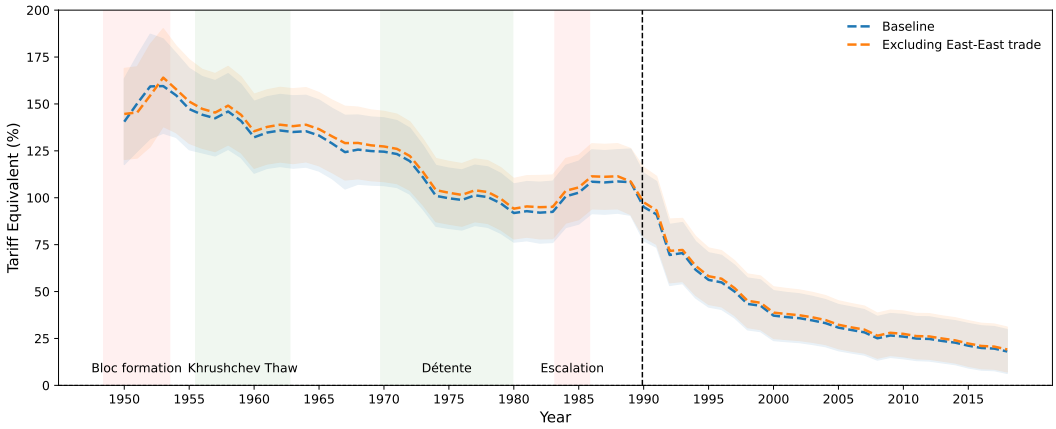


Figure D.4: Cross-bloc trade barrier: baseline vs. excluding East–East flows

Notes: The figure reports the estimated tariff-equivalent of the cross-bloc trade barrier under the baseline specification and a specification that excludes all international East–East flows. Both specifications are estimated using the same gravity equation with exporter-year and importer-year fixed effects and include domestic trade. In the exclusion sample, domestic trade for Eastern countries is retained to ensure identification (see Appendix C.3). Shaded areas represent 95% confidence intervals computed via the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2. The vertical dashed line marks the fall of the Berlin Wall.

D.4 Controlling for bilateral heterogeneity

A potential concern is that the estimated cross-bloc trade barrier may reflect time-invariant bilateral differences that are not fully captured by standard gravity controls such as distance and contiguity. For example, country pairs located on opposite sides of the Iron Curtain may differ systematically along dimensions that affect trade but are constant over time, such as differences in economic systems. Conversely, some cross-bloc pairs may share historical ties that have persistent effects on trade, such as the relationship between East and West Germany.

To address this issue, we replace the standard bilateral controls with country-pair fixed effects. we re-estimate the model by replacing standard bilateral controls with directed country-pair fixed effects, which absorb all persistent pair-specific factors affecting trade, including geographic, historical, and cultural factors. Because the inclusion of these fixed effects changes the normalization of the estimated coefficients, we rescale them to match the average level of the baseline estimates over the Cold-War sample period to facilitate comparison. This specification absorbs all time-invariant bilateral heterogeneity and identifies trade frictions exclusively from within-pair variation over time. Results are presented in Figure D.5.

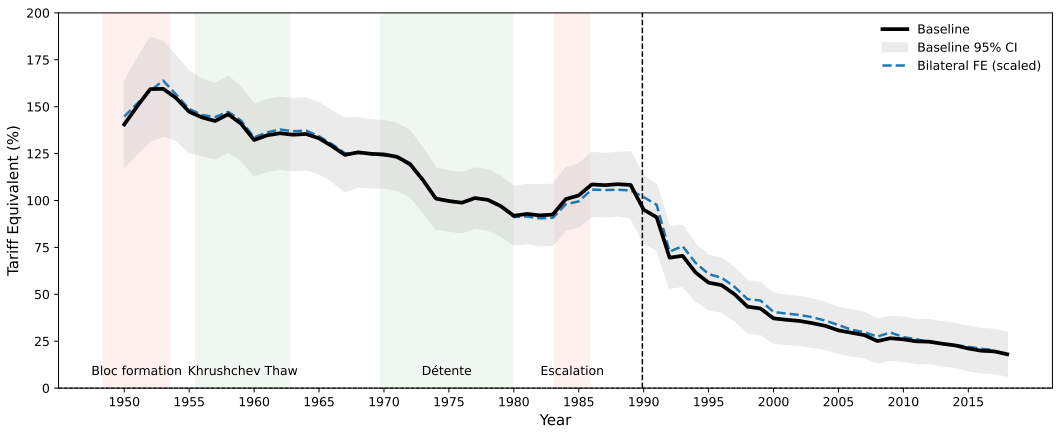


Figure D.5: Controlling for bilateral fixed effects: cross-bloc barrier

Notes: The figure reports the estimated tariff-equivalent of the cross-bloc trade barrier under the baseline specification and an alternative estimation in which time-varying controls are replaced by directed pair fixed effects. To ease the comparison, the coefficients of the fixed-effects estimation are scaled so that they have the same average as the baseline coefficients over the period 1950–1989. This transformation is done directly on the coefficients, before they are transformed into tariff equivalent measured. The shaded area represents the 95% confidence interval, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2. A vertical dashed line marks the fall of the Berlin Wall.

The results confirm that the main findings are robust to this specification. The estimated cross-bloc trade barrier displays a remarkably similar time profile, with declines during periods

of reduced geopolitical tension and increases during episodes of renewed conflict. While the level of the coefficients is not directly comparable, their evolution over time closely mirrors that of the baseline estimates. This indicates that our results are not driven by time-invariant bilateral differences, but instead reflect genuine time variation in trade frictions associated with geopolitical fragmentation.

These findings indicate that the main results are not driven by persistent bilateral differences between cross-bloc and within-bloc country pairs, but instead reflect genuine time variation in trade frictions associated with geopolitical fragmentation.

D.5 The influence of individual countries

A natural concern is that the estimated cross-bloc trade barrier may be driven by a small number of economically dominant or institutionally central countries. In the European context, Germany represents the most important bilateral relationship across the Cold War divide, while the Soviet Union accounts for a large share of trade within the Eastern bloc and may be subject to greater measurement uncertainty.

To assess the importance of these countries, we re-estimate the model excluding, in turn, (i) both East and West Germany, and (ii) the Soviet Union and its successor states. These exercises remove the most prominent cross-bloc bilateral relationship and the largest Eastern economy from the estimation sample. Results are presented in Figure D.6.

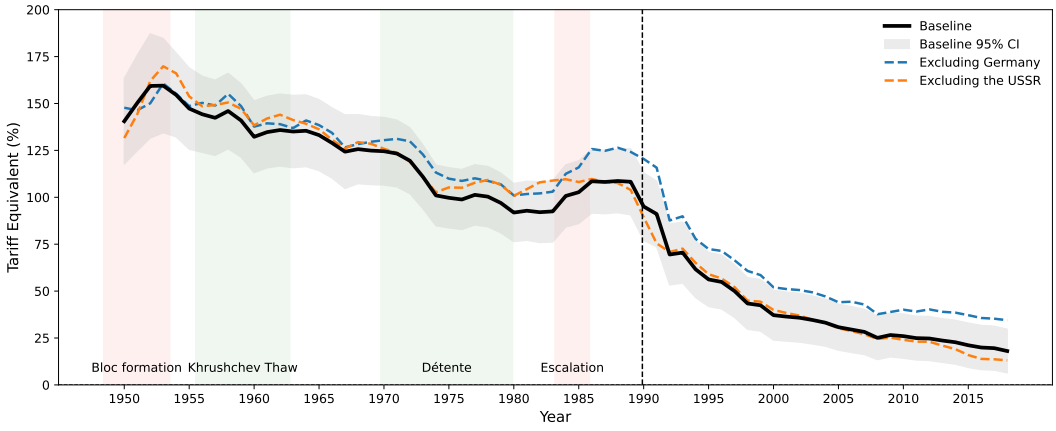


Figure D.6: Cross-bloc trade costs excluding key countries

Notes: The figure shows the estimated tariff equivalent of trade between countries located in different Cold War blocs (East–West), expressed in percentage points. The baseline series is compared to two alternative specifications that exclude, in turn, (i) both East and West Germany and (ii) the Soviet Union and its successor states. Estimates are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is computed as $100 \times [\exp(-\hat{\gamma}_t^{CB}/5) - 1]$, using a trade elasticity of 5. The shaded area represents a 95% confidence interval for the baseline estimate, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2, and the vertical dashed line marks the fall of the Berlin Wall.

The exclusion of Germany leads to a systematic shift in the estimated time profile of cross-bloc trade costs which is discussed in the main text in context of the onset of *Ostpolitik* and *Wandel durch Handel*.

By contrast, excluding the Soviet Union yields changes that are smaller and less systematic. Trade costs are somewhat higher in the early 1950s and during the later phase of Détente, and the secondary decline in trade costs after the plateau in the mid-1970s disappears. However,

these differences are modest relative to the baseline, and the overall time profile remains broadly unchanged.

Taken together, these results indicate that the main findings are not driven by any single country. At the same time, they highlight the importance of the German case in shaping the dynamics of cross-bloc trade during periods of geopolitical easing.

D.6 Alternative bloc definitions

A potential concern is that the results may depend on the composition of the sample or on the specific classification of countries into geopolitical blocs.

We first restrict the analysis to a core set of continental European countries, focusing on geographically central economies in both blocs. This restriction sharpens the comparison across the Iron Curtain by excluding peripheral countries whose trade patterns may be influenced by factors unrelated to geopolitical fragmentation.

Second, we consider alternative definitions of bloc membership based on reasonable variations in geopolitical alignment. These include reclassifying borderline cases and adjusting the composition of the Western bloc.

Across all variants, the results remain stable. The estimated cross-bloc trade barrier continues to display a large magnitude and a time pattern that closely follows major geopolitical phases. While changes in the sample affect the level of the estimated coefficients, the overall evolution of trade costs is preserved.

These findings indicate that the main results are not driven by peripheral countries or by specific choices in the classification of geopolitical blocs.

D.7 Directional asymmetries in cross-bloc trade

The baseline specification treats cross-bloc trade as symmetric, reflecting the fact that in a two-bloc setting with exporter-year and importer-year fixed effects, only the contrast between cross-bloc and within-bloc trade is identified. In particular, East-to-West and West-to-East trade costs cannot be separately estimated within the bloc-restricted sample.

To examine whether directional asymmetries in cross-bloc trade are empirically important, we extend the analysis by introducing a third group of countries. Specifically, we retain the European classification of Eastern and Western bloc countries and augment the sample with all remaining countries in the dataset, i.e., all countries outside the baseline European East–West classification, including countries that are not in Europe. These countries are grouped into a residual category ("Rest"), defined mechanically as those countries that are neither part of the European Eastern bloc nor the European Western bloc in our baseline classification. This additional variation is not absorbed by exporter-year and importer-year fixed effects and therefore allows East-to-West and West-to-East trade costs to be separately identified.

This approach comes with a tradeoff. While it increases the dimensionality of the identifying variation, it does so by introducing a heterogeneous set of countries that do not form a coherent geopolitical bloc. As a result, the identifying variation is no longer solely driven by the East–West partition that is central to our analysis. For this reason, we view this specification as a complementary extension rather than as part of the baseline empirical strategy.

We re-estimate the model allowing for separate coefficients for East-to-West and West-to-East trade. The results show no economically or statistically significant asymmetries between the two directions. This finding supports the symmetric treatment of cross-bloc trade in the baseline specification.

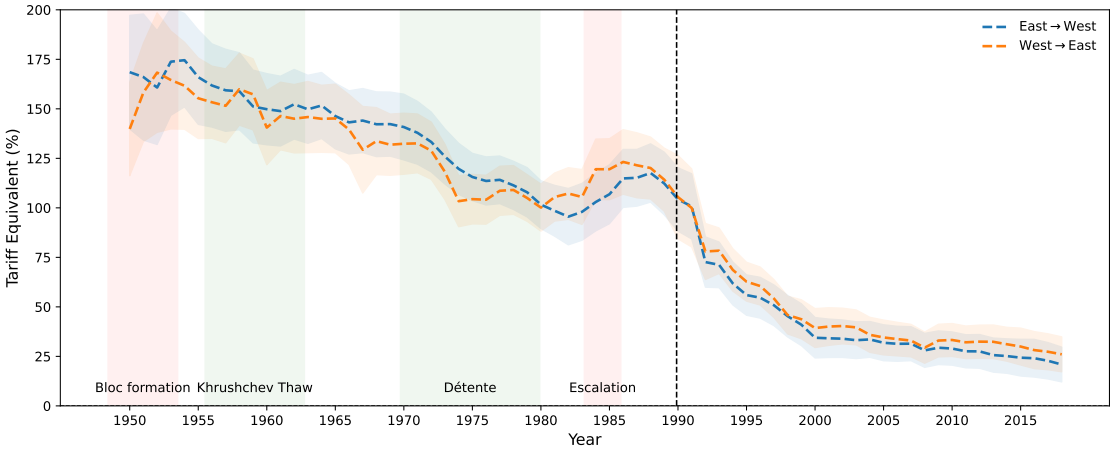


Figure D.7: Directional cross-bloc trade costs

Notes: The figure shows the estimated tariff equivalent of cross-bloc trade, allowing for directional asymmetries between East-to-West and West-to-East flows, expressed in percentage points. The estimates are obtained from specification (2) augmented with a third group of countries. The tariff equivalent is obtained by converting the estimated coefficients into ad valorem trade costs using a trade elasticity of 5. Shaded areas represent 95% confidence intervals, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2. A vertical dashed line marks the fall of the Berlin Wall.

D.8 Global sample: First and Second World

Our baseline analysis focuses on a bloc-restricted European sample, where the East–West division is historically sharp and provides clean identification. An important question is whether the patterns we document extend beyond Europe.

To address this, we replicate the analysis using a broader global sample. Specifically, we classify countries into geopolitical blocs based on the First and Second World distinction and estimate cross-bloc trade costs using the same empirical approach as in the baseline specification. The sample is restricted to countries in these two groups, yielding a setting that is directly comparable to the European baseline.

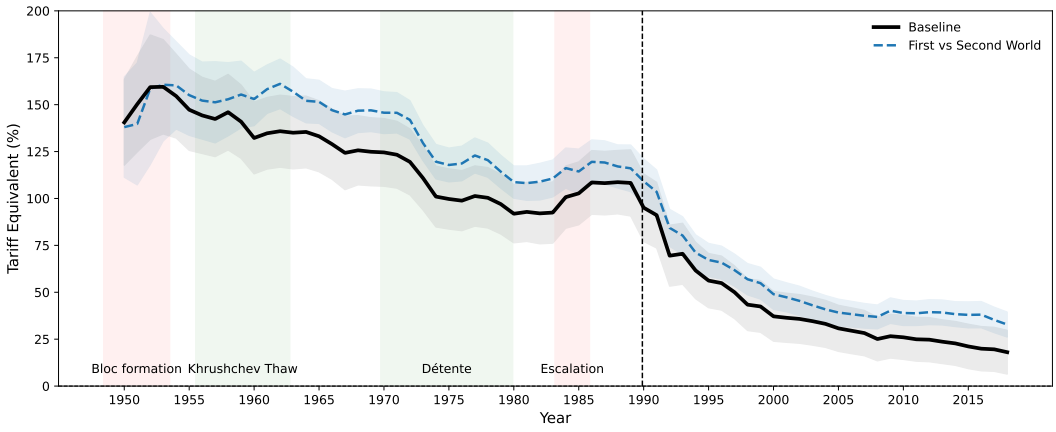


Figure D.8: Cross-bloc trade costs: global First vs Second World

Notes: The figure shows the estimated tariff equivalent of trade between countries classified as part of the First and Second World, expressed in percentage points. The sample is restricted to countries in these two groups. The estimates are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is obtained by converting the estimated coefficient $\hat{\gamma}_t^{CB}$ into an ad valorem trade cost using a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t^{CB}/5) - 1]$. The shaded area represents the 95% confidence interval, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2. A vertical dashed line marks the fall of the Berlin Wall.

Figure D.8 replicates the baseline analysis using a global sample restricted to countries classified as part of the First and Second World. The resulting estimates display a pattern that is similar to our baseline sample. Cross-bloc trade costs are large and decline over time, with movements that closely track major geopolitical phases of the Cold War. In particular, the global series captures the evolution of tensions during the Khrushchev Thaw, with trade costs peaking toward the end of this period, around the time of the Berlin Wall construction and the Cuban Missile Crisis.

Compared to the European baseline, the global estimates are in some periods even more tightly aligned with major geopolitical developments, reflecting the fact that these events were global in scope. At the same time, the underlying classification is less precise outside Europe, and countries do not fit as cleanly into a binary bloc structure. For this reason, we view the European sample as providing the most credible setting for identification, while the global analysis offers strong external validation of the main findings.

Comparison of data sources

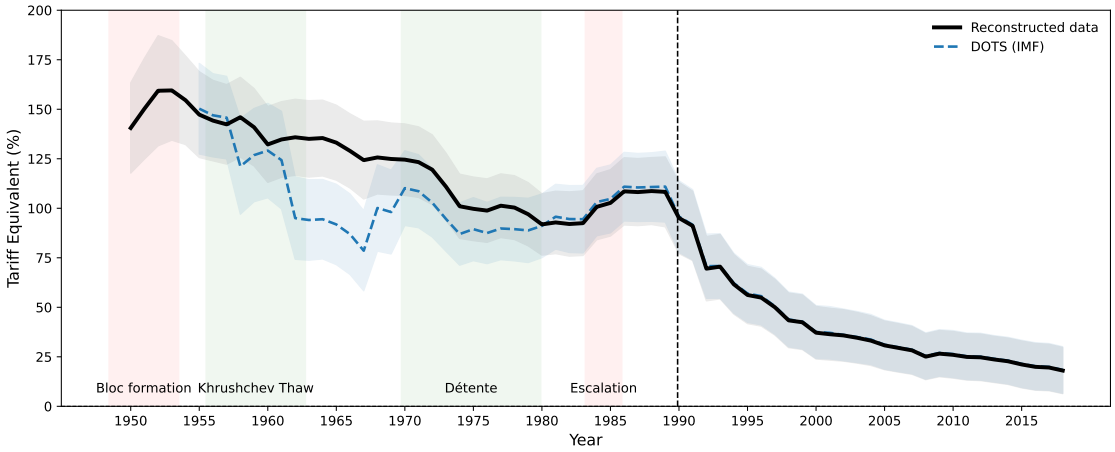


Figure D.9: Cross-bloc trade costs during the Cold War

Notes: The figure shows the estimated tariff equivalent of trade between countries located in different Cold War blocs (East–West), expressed in percentage points. The solid line reports estimates based on the reconstructed dataset, while the dashed line reports estimates based on the original IMF Direction of Trade Statistics (DOTS). Both series are obtained from specification (2), which includes domestic trade flows. The tariff equivalent is obtained by transforming the estimated coefficient $\hat{\gamma}_t^{CB}$ into an ad valorem trade cost using the standard gravity mapping with a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t^{CB}/5) - 1]$. The shaded area represents the 95% confidence interval, constructed using the delta method. Shaded bands indicate the main Cold War phases described in Section 2 and summarized in Table A.1.

D.9 Trade costs from symmetric bilateral measures

As a robustness check, we construct bilateral trade costs using the symmetric transformation proposed by Head and Ries (2001). Following Jacks et al. (2008), we aggregate these bilateral measures across country pairs to obtain time-series evidence on the evolution of trade costs across geopolitical groupings.

This approach relies on the same underlying gravity relationship as our baseline specification, but implements it through a transformation of observed trade flows rather than through the estimation of a full structural model. It yields direct bilateral measures of trade costs that can be aggregated across country pairs to trace the evolution of trade frictions across geopolitical groupings.

Under a standard Armington gravity model with iceberg trade costs and trade elasticity $\theta > 0$, bilateral trade flows satisfy

$$X_{ijt} = \frac{Y_{it}E_{jt}}{Y_t}\left(\frac{\tau_{ijt}}{\Pi_{it}P_{jt}}\right)^{-\theta}.$$

Assuming symmetric trade costs, $\tau_{ijt} = \tau_{jit}$, and normalizing domestic trade costs to one, $\tau_{iit} = 1$, yields

$$y_{ijt} \equiv \frac{1}{2}\left[\log X_{iit} + \log X_{jjt} - \log X_{ijt} - \log X_{jit}\right] = \theta \log \tau_{ijt}.$$

This transformation differences out country size, expenditure, and multilateral resistance terms and can be interpreted as comparing the intensity of bilateral trade to domestic trade in both directions. Bilateral trade costs can then recovered as $\tau_{ijt} = \exp(y_{ijt}/\theta)$, using the same trade elasticity as in the baseline analysis.

We construct y_{ijt} for each unordered country pair and use it to obtain a direct measure of bilateral trade costs and their tariff-equivalent expression. To summarize these bilateral measures over time, we group country pairs according to their geopolitical alignment and compute average trade costs within each group. In particular, we distinguish between within-West (WW), within-East (EE), and cross-bloc (CB) pairs, and construct a time series of average trade costs for each category. The resulting series for cross-bloc pairs provides a direct counterpart to the cross-bloc trade-cost measure reported in Figure 3.

An important distinction relative to the baseline approach is that the symmetric transformation removes exporter-time and importer-time components but does not eliminate time-invariant bilateral heterogeneity. The transformed measure therefore retains persistent pair-specific components related to geography, such as distance and contiguity.

This measure offers several advantages. It is simple and transparent, and it relies only on observed

bilateral and domestic trade flows, without requiring the estimation of high-dimensional fixed effects. At the same time, it is more restrictive than structural gravity specifications. It imposes symmetry of bilateral trade costs and therefore cannot distinguish between East-to-West and West-to-East trade barriers, even with the addition of a third comparison group. More importantly, it excludes country pairs with zero trade flows.

In addition, measurement error is likely to be amplified in this framework. The bilateral trade-cost measure combines four trade-flow observations for each country pair, so that noise in any of these components propagates nonlinearly into the transformed variable. This issue is particularly relevant in our historical setting, where both bilateral trade flows and domestic trade are subject to substantial measurement uncertainty. Moreover, because the transformation is applied prior to aggregation, measurement error at the bilateral level is not averaged out in a structured way and may instead bias group-level comparisons. By contrast, the structural gravity specification aggregates information across all bilateral flows within a coherent estimation framework and is better suited to handle missing observations, zero trade flows, and heterogeneous data quality.

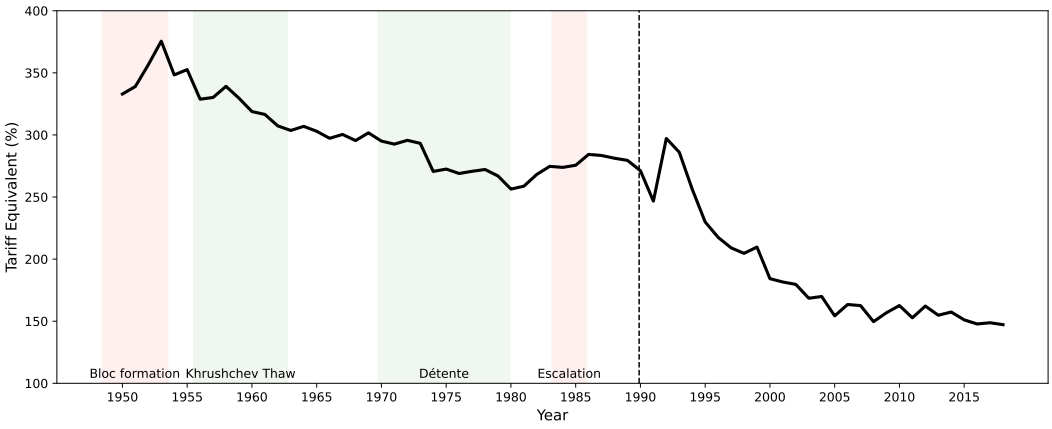


Figure D.10: Cross-bloc trade costs from the Head–Ries transformation

Notes: The figure shows the tariff-equivalent measure of trade costs for cross-bloc country pairs obtained from the symmetric Head–Ries transformation (Head and Ries, 2001), expressed in percentage points. For each unordered country pair, the bilateral trade-cost measure is constructed from bilateral and domestic trade flows under the assumption of symmetric bilateral frictions. Country pairs are then grouped according to bloc membership, and the figure reports the average trade-cost measure for cross-bloc pairs over time. The tariff equivalent is obtained by converting the estimated bilateral trade-cost measure into an ad valorem trade cost using a trade elasticity of 5. Shaded bands indicate the main Cold War phases discussed in Section 2. A vertical dashed line marks the fall of the Berlin Wall.

The resulting series confirms the presence of a substantial cross-bloc trade wedge during the Cold War and a marked decline after its end. The time profile broadly aligns with the baseline

results prior to the fall of the Berlin Wall, with similar turning points and responses to major geopolitical events. However, the series displays a sharp temporary increase following the collapse of the Eastern bloc.

This difference reflects how the two approaches use the underlying data. The bilateral transformation constructs trade-cost measures at the country-pair level, each of which depends on four trade-flow observations. As a result, measurement error in bilateral or domestic trade flows is directly propagated into the transformed variable. In our historical setting, where both types of flows are subject to non-negligible measurement uncertainty, this can lead to relatively noisy estimates when aggregating across country pairs. By contrast, the structural gravity specification combines information from all bilateral observations within a unified estimation framework, which helps to mitigate the influence of measurement error and yields more stable estimates of cross-bloc trade costs over time.

The symmetric bilateral measure thus confirm the presence of a substantial cross-bloc trade wedge, while underscoring the advantages of the structural gravity framework for quantifying its magnitude and evolution over time.

D.10 Neutral Countries: Classification and Empirical Design

Country classification

The classification of countries as neutral is based on their geopolitical position during the Cold War, drawing on historical and institutional criteria. Our objective is to identify countries that were not formally part of either bloc while maintaining a classification that is both economically meaningful and stable over time.

We classify Austria, Switzerland, Sweden, and Ireland as neutral countries, reflecting their formal political neutrality and absence from bloc institutions. Finland and Yugoslavia are classified as non-aligned countries. Finland combined formal neutrality with significant geopolitical constraints in its relationship with the Soviet Union, while Yugoslavia followed an independent socialist path outside the Soviet sphere and maintained economic relations with both blocs.

We exclude Albania, despite its potential classification as non-aligned, due to its limited integration into international trade and its idiosyncratic geopolitical trajectory, which would add noise without providing meaningful identifying variation. Throughout the analysis, these classifications are treated as time-invariant to maintain a consistent mapping between geopolitical alignment and trade patterns.

Results for non-aligned countries are reported separately and are less conclusive, reflecting the small and heterogeneous nature of this group.

Empirical specification

We extend the baseline specification in equation (2) by incorporating countries that were not formally part of either bloc. Let N_i denote an indicator for whether country i is neutral. We define bilateral indicators for trade between neutral countries and each bloc:

$$EN_{ij} \equiv \mathbf{1}\{i \in East,\ j \in Neutral\} + \mathbf{1}\{i \in Neutral,\ j \in East\},$$

$$WN_{ij} \equiv \mathbf{1}\{i \in West,\ j \in Neutral\} + \mathbf{1}\{i \in Neutral,\ j \in West\},$$

as well as an indicator for trade among neutral countries:

$$NN_{ij} \equiv \mathbf{1}\{i \in Neutral,\ j \in Neutral,\ i \neq j\}.$$

We estimate the following augmented gravity equation:

$$X_{ijt} = \exp\Big(\gamma_t^{CB}\, CB_{ij} + \gamma_t^{EE}\, EE_{ij} + \gamma_t^{WW}\, WW_{ij} + \gamma_t^{EN}\, EN_{ij} + \gamma_t^{WN}\, WN_{ij} + \gamma_t^{NN}\, NN_{ij} + \phi_{it} + \psi_{jt} + z_{ij}'\beta\Big) + \varepsilon_{ijt}, \tag{D.1}$$

where the sample includes both international and domestic trade flows, and domestic trade ($i = j$) is the omitted category.

As in the baseline specification, coefficients are identified from bilateral variation conditional on exporter-year and importer-year fixed effects. All indicators are defined symmetrically, so that directional asymmetries are not separately identified. The coefficients of primary interest are γ_t^{EN} and γ_t^{WN}, which capture trade frictions between neutral countries and each bloc relative to domestic trade.

Identification and interpretation

Including neutral countries expands the set of bilateral comparisons relative to the bloc-restricted sample. In addition to the East–East, West–West, and East–West contrasts, the specification exploits variation in trade between neutral countries and each bloc, as well as among neutral countries themselves.

Conditional on exporter-year and importer-year fixed effects, identification continues to rely on how trade is allocated across partners within a country-year. The inclusion of domestic trade provides the benchmark that allows all bilateral coefficients to be interpreted as border effects relative to trading at home. The coefficients γ_t^{EN} and γ_t^{WN} therefore measure the trade wedges associated with trading between neutral countries and each bloc, relative to domestic trade.

As in the baseline model, the interpretation of these coefficients depends on the assumption that domestic trade provides a comparable benchmark across countries and over time. To the extent that domestic trade is measured with error or evolves differently across groups, this may affect the level of the estimated coefficients. However, the time variation remains informative as long as such differences are relatively stable over time.

D.11 European non-aligned countries

This subsection reports results for European non-aligned countries, focusing on Finland and Yugoslavia. Let A_i denote an indicator for whether country i is non-aligned. These countries were not formally part of either bloc but maintained more complex and often constrained relationships with both sides.

We define bilateral indicators for trade between non-aligned countries and each bloc:

$$EA_{ij} \equiv \mathbf{1}\{i \in East,\ j \in NonAligned\} + \mathbf{1}\{i \in NonAligned,\ j \in East\},$$

$$WA_{ij} \equiv \mathbf{1}\{i \in West,\ j \in NonAligned\} + \mathbf{1}\{i \in NonAligned,\ j \in West\},$$

as well as an indicator for trade among non-aligned countries, AA_{ij}.

We estimate a specification that is analogous to the specification that includes neutral countries, but where neutral countries are replaced with non-aligned countries:

$$\begin{aligned} X_{ijt} = \exp\Big(&\gamma_t^{CB}\, CB_{ij} + \gamma_t^{EE}\, EE_{ij} + \gamma_t^{WW}\, WW_{ij} \\ &+ \gamma_t^{EA}\, EA_{ij} + \gamma_t^{WA}\, WA_{ij} + \gamma_t^{AA}\, AA_{ij} \\ &+ \phi_{it} + \psi_{jt} + z_{ij}'\beta\Big) + \varepsilon_{ijt}. \end{aligned}$$

The coefficients of interest in this case are γ_t^{EA} and γ_t^{WA}.

Figure D.11 reports the estimated trade costs for non-aligned countries when trading with the Western and Eastern blocs. The results indicate that these countries occupy an intermediate position between the two blocs, with a persistent tilt toward the West.

Trade costs between non-aligned countries and the Western bloc are higher than those observed for neutral countries, indicating a lower degree of integration with Western economies. At the same time, trade costs with the Eastern bloc are also elevated, suggesting that non-aligned countries were not fully integrated into either system. Despite this, trade costs with the West remain consistently below those with the East, pointing to a persistent orientation toward Western markets.

A notable feature of the data is that, during the first half of the 1980s, trade costs with the East and the West become very similar. This pattern suggests a period in which non-aligned countries faced comparable frictions when trading with both blocs, consistent with a more balanced or constrained position during a phase of heightened geopolitical tension.

In contrast to the results for neutral countries, the time variation in trade costs with the Eastern bloc is less closely aligned with major geopolitical phases. While some response to events is

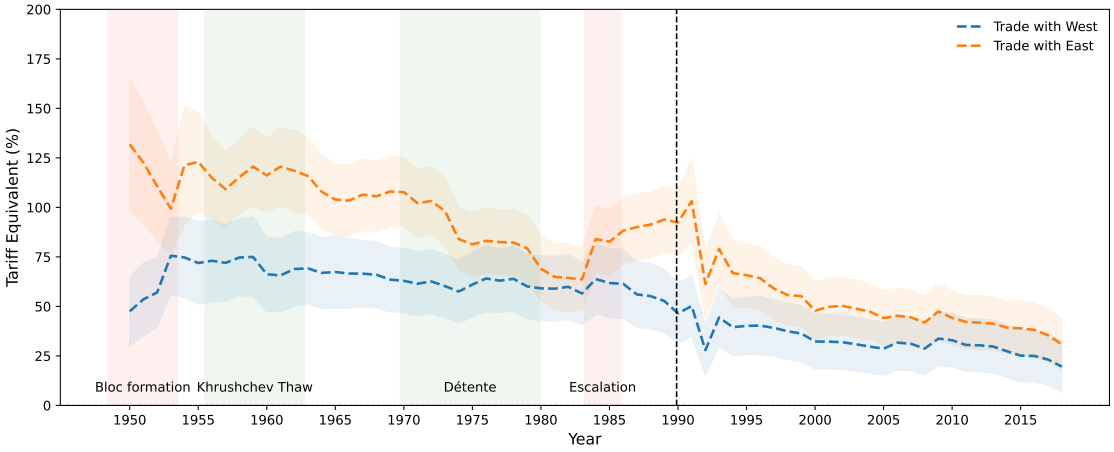


Figure D.11: Trade costs for non-aligned countries

Notes: The figure shows the estimated tariff equivalent of trade between non-aligned countries and the Eastern and Western blocs, expressed in percentage points. The estimates are obtained from an augmented version of specification (2) that includes non-aligned countries and bilateral indicators for trade between non-aligned countries and each bloc. The tariff equivalent is obtained by converting the estimated coefficients into ad valorem trade costs using a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t/5) - 1]$. The shaded areas represent 95% confidence intervals, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2, and the vertical dashed line marks the fall of the Berlin Wall. The non-aligned category includes Finland and Yugoslavia; estimates should therefore be interpreted with caution.

visible, it is more muted, indicating that trade patterns for non-aligned countries were less tightly linked to the evolution of East–West relations.

These findings should be interpreted with caution. The non-aligned category in our sample consists of only two countries, Finland and Yugoslavia, which followed markedly different economic and geopolitical trajectories. Finland maintained formal neutrality but became increasingly integrated with Western Europe over time, while Yugoslavia pursued an independent socialist path outside the Soviet sphere. As a result, the estimated coefficients reflect an average of heterogeneous experiences rather than a unified pattern of behavior.

Taken together, these results suggest that non-aligned countries occupied an intermediate and less clearly defined position in the Cold War trading system, characterized by partial integration with both blocs and substantial heterogeneity across countries.

We further extend the analysis to a global setting by focusing on a core group of extra-European non-aligned countries with sustained participation in the Non-Aligned Movement.

D.12 Global non-aligned countries

In this subsection, we consider a broader set of non-aligned countries in a global setting. Specifically, we focus on a core group of countries with sustained participation in the Non-Aligned Movement, consisting of India, Indonesia, Egypt, Algeria, Ghana, and Tanzania. This selection captures a set of economies that were not formally aligned with either bloc but maintained economic relations with both sides. The specification used is the same as the one used in the previous subsection.

Figure D.12 reports the estimated trade costs between these countries and the Eastern and Western blocs. The results indicate that non-aligned countries occupy an intermediate position between the two blocs throughout the sample period. Trade costs with with the Eastern bloc are initially high, particularly in the early 1950s while those with the Wetern bloc start from a lower level.

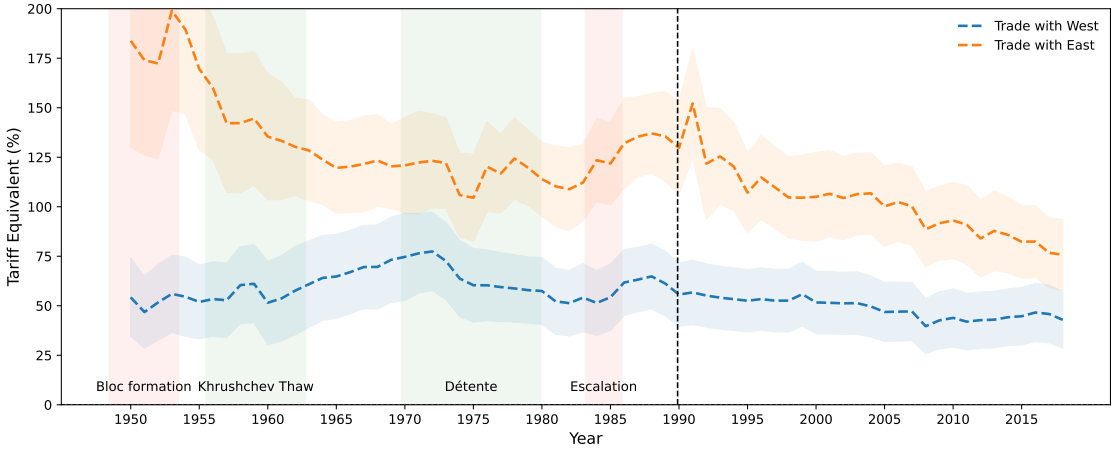


Figure D.12: Trade costs for global non-aligned countries

Notes: The figure shows the estimated tariff equivalent of trade between a core group of non-aligned countries and the Eastern and Western blocs, expressed in percentage points. The non-aligned group consists of India, Indonesia, Egypt, Algeria, Ghana, and Tanzania. The estimates are obtained from an augmented version of specification (2) that includes these countries and bilateral indicators for trade between non-aligned countries and each bloc. The tariff equivalent is obtained by converting the estimated coefficients into ad valorem trade costs using a trade elasticity of 5, i.e., $100 \times [\exp(-\hat{\gamma}_t/5) - 1]$. The shaded areas represent 95% confidence intervals, constructed using the delta method. Shaded bands indicate the main Cold War phases discussed in Section 2, and the vertical dashed line marks the fall of the Berlin Wall.

Over time, however, trade costs with the two blocs evolve asymmetrically. Trade costs with the Eastern bloc decline steadily from very high initial levels, while trade costs with the Western

bloc increase from relatively low levels before stabilizing. As a result, trade costs with the two blocs converge over time, indicating a shift from an initially asymmetric configuration toward a more balanced trading relationship.

In addition to this convergence, the time variation in trade costs displays co-movement after the period of Détente when trade costs with both blocs increase again during the renewed escalation of tensions in the early 1980s. This pattern suggests that geopolitical conditions may have affected trade relationships not only across blocs but also for countries outside the formal bloc structure.

At the same time, the interpretation of these results requires caution. The global sample spans countries with different economic structures, geographic characteristics, and historical trajectories, and therefore does not provide the same tightly defined setting as the European analysis. For this reason, we view these results as providing complementary evidence on the role of geopolitical fragmentation, rather than as a direct counterpart to the European findings.

E Quantitative Model with Decentralized West and Centralized East

E.1 Welfare effects

To quantify the consequences of geopolitical fragmentation, we use a structural model that maps observed changes in bilateral trade patterns into an implied cross-bloc trade-cost change, $\hat{\tau}^{CB}$, and then evaluates the corresponding welfare effects. We focus on two transitions that capture the main turning points in the evolution of Cold War trade frictions: the move from the early peak of fragmentation to the post-Détente trough, and the subsequent reversal from the post-Détente trough to the late peak. These two episodes isolate, respectively, a major decline in cross-bloc barriers and a later episode of renewed fragmentation.

As explained in more detail below, the quantitative results depend on three elasticities that govern how trade patterns respond to changes in trade costs. The first is the standard Armington trade elasticity, which controls substitution across all origins in market economies. We assume that trade into Western destinations is governed by this elasticity and set it to the conventional value $\theta_W = 5$. For Eastern destinations, by contrast, θ_E governs substitution across foreign suppliers within the import procurement system, while ρ_E governs substitution between domestic absorption and the aggregate import bundle. This distinction is meant to capture the institutional organization of trade in the Eastern bloc, where centralized procurement and planning limited the responsiveness of trade flows to relative prices. Our baseline case sets $(\rho_E, \theta_E, \theta_W) = (2, 3, 5)$ and therefore combines a conventional Western elasticity with more limited substitution on the Eastern side. To assess robustness, Table E.1 also reports a reference case, $(5, 5, 5)$, in which the asymmetry between Eastern and Western substitution patterns is removed. Comparing these two specifications allows us to distinguish results that depend on the distinctive institutional features of Eastern trade from those that arise more generally from the evolution of cross-bloc frictions.

Table E.1 reports the estimated cross-bloc trade-cost changes and the associated welfare effects for these two benchmark transitions. The estimated trade-cost movements line up closely with the historical narrative. In the baseline specification, the move from the early peak to the post-Détente trough implies $\hat{\tau}^{CB} = 0.669$, while the reversal from the trough to the late peak implies $\hat{\tau}^{CB} = 1.187$. The reference case produces the same qualitative pattern, with a somewhat smaller decline in the first transition and a somewhat smaller increase in the second. The model therefore recovers a large fall in fragmentation during the period of East-West relaxation and a more moderate re-tightening during the later phase of renewed tensions.

The welfare effects are economically meaningful and strongly asymmetric across blocs. Evaluated

Table E.1: Cross-bloc trade costs and welfare effects of geopolitical fragmentation

Transition	$(\rho_E, \theta_E, \theta_W)$	$\hat{\tau}^{CB}$	**West**	**East**
Early peak → Trough	$(2, 3, 5)$	0.669	0.71 [0.21, 1.26]	2.96 [1.40, 5.05]
	$(5, 5, 5)$	0.781	0.68 [0.21, 1.21]	1.50 [0.66, 2.62]
Trough → Late peak	$(2, 3, 5)$	1.187	-0.15 [-0.26, -0.04]	-0.60 [-0.98, -0.27]
	$(5, 5, 5)$	1.114	-0.14 [-0.26, -0.04]	-0.31 [-0.55, -0.14]

Notes: The table reports the estimated cross-bloc trade-cost change, $\hat{\tau}^{CB}$, and the associated welfare effects for two benchmark transitions: from the early peak of fragmentation (1951–1953) to the post-Détente trough (1980–1982), and from the post-Détente trough (1980–1982) to the late peak (1986–1988). Welfare is evaluated on the common post-Détente trough trade structure. Entries in the West and East columns report average welfare changes, in percent, across Western and Eastern countries, respectively; the bracketed values indicate the minimum and maximum country-level welfare changes within each bloc. The baseline elasticity configuration is $(\rho_E, \theta_E, \theta_W) = (2, 3, 5)$; the reference case is $(5, 5, 5)$.

on the common post-Détente trough trade structure, the move from the early peak to the trough raises average welfare in the West by about 0.7 percent, but raises average welfare in the East by about 3.0 percent in the baseline case. Conversely, the increase in fragmentation from the trough to the late peak lowers average welfare in the West by roughly 0.15 percent, compared with about 0.60 percent in the East. The reference case yields somewhat smaller effects for the East, but preserves the same qualitative ranking of episodes and the same asymmetry across blocs. The main implication is therefore robust: changes in cross-bloc trade frictions mattered much more for Eastern economies than for Western ones, so that the East gained disproportionately from geopolitical relaxation and bore disproportionately larger costs when fragmentation intensified again.

E.2 Model environment

We consider a one-sector general-equilibrium trade model with two types of destinations. Eastern countries are modeled as economies in which foreign trade is intermediated by a centralized state trading agency. Western countries are modeled as decentralized market economies with standard CES demand over origins.

Let the set of countries be partitioned into Eastern countries, E, and Western countries, W. Country i produces a differentiated good and ships it to destination j subject to iceberg trade costs $\tau_{ij} \geq 1$. Let p_i denote the producer price in country i, so that the delivered price in destination j is

$$q_{ij} = p_i \tau_{ij}. \tag{E.1}$$

E.3 Eastern destinations

For each Eastern destination $j \in E$, final absorption is produced by combining the domestic good with an aggregate of imported goods. The key institutional assumption is that imports are purchased by a centralized state trading agency. The agency is modeled as a cost-minimizing procurement institution: it chooses foreign-source quantities to produce a given import aggregate, taking foreign delivered prices as given.

Lower-level import aggregation. The state trading agency produces the import aggregate

$$m_j = \left[\sum_{i \neq j} (\alpha_{ij}^E)^{1/\theta_E} x_{ij}^{\frac{\theta_E - 1}{\theta_E}} \right]^{\frac{\theta_E}{\theta_E - 1}}, \qquad j \in E, \tag{E.2}$$

where x_{ij} is the physical quantity imported from origin i, $\alpha_{ij}^E > 0$ is a source-specific weight, and $\theta_E > 1$ is the elasticity of substitution across foreign suppliers. The weights are normalized so that

$$\sum_{i \neq j} \alpha_{ij}^E = 1, \qquad j \in E. \tag{E.3}$$

The agency solves

$$\min_{\{x_{ij}\}_{i \neq j}} \sum_{i \neq j} p_i \tau_{ij} x_{ij} \quad \text{subject to} \quad m_j = \left[\sum_{i \neq j} (\alpha_{ij}^E)^{1/\theta_E} x_{ij}^{\frac{\theta_E - 1}{\theta_E}} \right]^{\frac{\theta_E}{\theta_E - 1}}. \tag{E.4}$$

The associated unit cost of the import aggregate is

$$P_j^M = \left[\sum_{i \neq j} \alpha_{ij}^E (p_i \tau_{ij})^{1 - \theta_E} \right]^{\frac{1}{1 - \theta_E}}, \qquad j \in E, \tag{E.5}$$

and bilateral import expenditure is

$$X_{ij} = \alpha_{ij}^E \left(\frac{p_i \tau_{ij}}{P_j^M} \right)^{1 - \theta_E} M_j, \qquad j \in E, \quad i \neq j, \tag{E.6}$$

where $M_j \equiv P_j^M m_j$ denotes total expenditure on imports.

Upper-level domestic-import allocation. Final absorption in Eastern country j is produced by combining the domestic good and the import aggregate:

$$C_j = \left[\beta_j^{1/\rho_E} d_j^{\frac{\rho_E-1}{\rho_E}} + (1-\beta_j)^{1/\rho_E} m_j^{\frac{\rho_E-1}{\rho_E}}\right]^{\frac{\rho_E}{\rho_E-1}}, \qquad j \in E, \tag{E.7}$$

where d_j is domestic absorption of the home good, so that $X_{jj} = p_j d_j$, $\beta_j \in (0,1)$ is a domestic-bias parameter, and $\rho_E > 1$ is the elasticity of substitution between domestic goods and the import aggregate. In the case $\rho_E = 1$, the CES function is replaced with the usual Cobb-Douglas aggregator.

The upper-level cost-minimization problem is

$$\min_{d_j, m_j} p_j d_j + P_j^M m_j \quad \text{subject to} \quad C_j = \left[\beta_j^{1/\rho_E} d_j^{\frac{\rho_E-1}{\rho_E}} + (1-\beta_j)^{1/\rho_E} m_j^{\frac{\rho_E-1}{\rho_E}}\right]^{\frac{\rho_E}{\rho_E-1}}. \tag{E.8}$$

The corresponding Eastern absorption price index is

$$P_j^E = \left[\beta_j p_j^{1-\rho_E} + (1-\beta_j)(P_j^M)^{1-\rho_E}\right]^{\frac{1}{1-\rho_E}}, \qquad j \in E. \tag{E.9}$$

Expenditure on the domestic good and on the import aggregate satisfies

$$X_{jj} = \beta_j \left(\frac{p_j}{P_j^E}\right)^{1-\rho_E} \mathcal{E}_j, \qquad j \in E, \tag{E.10}$$

$$M_j = (1-\beta_j) \left(\frac{P_j^M}{P_j^E}\right)^{1-\rho_E} \mathcal{E}_j, \qquad j \in E, \tag{E.11}$$

where $\mathcal{E}_j$ is total expenditure.

Thus, for Eastern destinations, substitution across foreign suppliers is governed by θ_E, while substitution between domestic goods and the import aggregate is governed by ρ_E.

E.4 Western destinations

For each Western destination $j \in W$, final expenditure is allocated across all origins according to a standard CES aggregator with elasticity $\theta_W > 1$:

$$C_j^W = \left[\sum_i (\alpha_{ij}^W)^{1/\theta_W} c_{ij}^{\frac{\theta_W-1}{\theta_W}}\right]^{\frac{\theta_W}{\theta_W-1}}, \qquad j \in W, \tag{E.12}$$

where c_{ij} is the physical quantity purchased from origin i and $\alpha_{ij}^W > 0$ is an origin-specific weight. The weights are normalized so that

$$\sum_i \alpha_{ij}^W = 1, \qquad j \in W. \tag{E.13}$$

Cost minimization implies the Western price index

$$P_j^W = \left[\sum_i \alpha_{ij}^W (p_i \tau_{ij})^{1-\theta_W}\right]^{\frac{1}{1-\theta_W}}, \qquad j \in W, \tag{E.14}$$

and bilateral expenditure

$$X_{ij} = \alpha_{ij}^W \left(\frac{p_i \tau_{ij}}{P_j^W}\right)^{1-\theta_W} \mathcal{E}_j, \qquad j \in W. \tag{E.15}$$

E.5 Equilibrium

The model takes as given trade costs $\{\tau_{ij}\}$, preference parameters $\{\alpha_{ij}^E, \alpha_{ij}^W, \beta_j\}$, elasticities θ_E, ρ_E, and θ_W, and exogenous output levels $\{\bar{y}_i\}$.

An equilibrium consists of prices $\{p_i\}$, expenditures $\{\mathcal{E}_j\}$, income levels $\{Y_i\}$, and trade flows $\{X_{ij}\}$ such that:

1. Eastern allocations satisfy (E.6), (E.10), and (E.11), with price indices given by (E.5) and (E.9).
2. Western allocations satisfy (E.15), with price indices given by (E.14).
3. Goods markets clear:
$$Y_i = \sum_j X_{ij}. \tag{E.16}$$
4. Nominal income equals the value of production:
$$Y_i = p_i \bar{y}_i. \tag{E.17}$$
5. Expenditure is proportional to income:
$$\mathcal{E}_j = \xi_j Y_j, \quad \xi_i > 0. \tag{E.18}$$

Preference parameters are calibrated to match observed bilateral trade shares in the baseline and are held fixed in counterfactuals. In the exact-hat implementation below, the calibrated preference parameters enter only through baseline expenditure shares.

E.6 Calibration and baseline shares

For each origin i, define the baseline sales shares

$$r_{ij} \equiv \frac{X_{ij}}{Y_i}, \qquad \sum_j r_{ij} = 1. \tag{E.19}$$

For each Western destination $j \in W$, define baseline expenditure shares

$$\pi_{ij}^W \equiv \frac{X_{ij}}{\mathcal{E}_j}, \qquad \sum_i \pi_{ij}^W = 1. \tag{E.20}$$

For each Eastern destination $j \in E$, define the baseline domestic expenditure share

$$s_{jj} \equiv \frac{X_{jj}}{\mathcal{E}_j}, \tag{E.21}$$

the baseline import expenditure share

$$\mu_j \equiv \frac{M_j}{\mathcal{E}_j} = 1 - s_{jj}, \tag{E.22}$$

and the baseline foreign-source shares within the import bundle

$$\lambda_{ij} \equiv \frac{X_{ij}}{M_j}, \qquad i \neq j, \qquad \sum_{i \neq j} \lambda_{ij} = 1. \tag{E.23}$$

These imply

$$\frac{X_{ij}}{\mathcal{E}_j} = \mu_j \lambda_{ij}, \qquad i \neq j, \quad j \in E. \tag{E.24}$$

E.7 Exact-hat algebra

Let hats denote counterfactual-to-baseline ratios:

$$\widehat{z} \equiv \frac{z'}{z}. \tag{E.25}$$

With fixed output, (E.17) implies

$$\widehat{Y}_i = \widehat{p}_i. \tag{E.26}$$

From (E.18),

$$\widehat{\mathcal{E}}_j = \widehat{Y}_j. \tag{E.27}$$

Eastern destinations. For each $j \in E$, the lower-level import price index is

$$\widehat{P}_j^M = \left[\sum_{i \neq j} \lambda_{ij} (\widehat{p}_i \widehat{\tau}_{ij})^{1-\theta_E}\right]^{\frac{1}{1-\theta_E}}. \tag{E.28}$$

The upper-level absorption price index is

$$\widehat{P}_j^E = \left[s_{jj} \widehat{p}_j^{1-\rho_E} + (1 - s_{jj})(\widehat{P}_j^M)^{1-\rho_E}\right]^{\frac{1}{1-\rho_E}}. \tag{E.29}$$

Import expenditure satisfies

$$\widehat{M}_j = \left(\frac{\widehat{P}_j^M}{\widehat{P}_j^E}\right)^{1-\rho_E} \widehat{\mathcal{E}}_j. \tag{E.30}$$

For foreign origins $i \neq j$, bilateral import expenditure satisfies

$$\widehat{X}_{ij} = \left(\frac{\widehat{p}_i \widehat{\tau}_{ij}}{\widehat{P}_j^M}\right)^{1-\theta_E} \widehat{M}_j, \qquad j \in E, \quad i \neq j. \tag{E.31}$$

Domestic trade in Eastern country j satisfies

$$\widehat{X}_{jj} = \left(\frac{\widehat{p}_j}{\widehat{P}_j^E}\right)^{1-\rho_E} \widehat{\mathcal{E}}_j, \qquad j \in E. \tag{E.32}$$

Western destinations. For each $j \in W$, the price index satisfies

$$\widehat{P}_j^W = \left[\sum_i \pi_{ij}^W (\widehat{p}_i \widehat{\tau}_{ij})^{1-\theta_W}\right]^{\frac{1}{1-\theta_W}}, \tag{E.33}$$

and bilateral expenditure satisfies

$$\widehat{X}_{ij} = \left(\frac{\widehat{p}_i \widehat{\tau}_{ij}}{\widehat{P}_j^W}\right)^{1-\theta_W} \widehat{\mathcal{E}}_j, \qquad j \in W. \tag{E.34}$$

Market clearing. Using baseline sales shares, exact-hat market clearing is

$$\widehat{Y}_i = \sum_j r_{ij} \widehat{X}_{ij}. \tag{E.35}$$

E.8 Counterfactual implementation

The counterfactual is implemented by changing bilateral trade costs while holding fixed the calibrated preference parameters and elasticities. In the application, the estimated time-varying geopolitical barrier is mapped into changes in $\widehat{\tau}_{ij}$ for cross-bloc country pairs:

$$\widehat{\tau}_{ij} = \begin{cases} \widehat{b}_t, & \text{if } i \in W,\ j \in E, \\ \widehat{b}_t, & \text{if } i \in E,\ j \in W, \\ 1, & \text{otherwise.} \end{cases} \tag{E.36}$$

Domestic trade and within-bloc trade costs are held fixed.

Solution algorithm. The equilibrium is solved by iterating on the vector of producer-price changes $\{\widehat{p}_i\}_i$.

Given an initial guess for $\{\widehat{p}_i\}_i$:

1. Use (E.28), (E.29), and (E.33) to compute the Eastern import price indices, Eastern absorption price indices, and Western price indices.
2. Use (E.27) together with the current income vector to compute $\{\widehat{\mathcal{E}}_j\}_j$.
3. Use (E.30), (E.31), (E.32), and (E.34) to compute bilateral trade-flow changes.
4. Use market clearing, (E.35), to update income changes $\{\widehat{Y}_i\}_i$.
5. Use the fixed-output relation (E.26) to update producer-price changes.

The procedure is repeated until convergence.

E.9 Welfare

Welfare is measured by real absorption. Under the expenditure-income closure in (E.18), proportional changes in real absorption coincide with proportional changes in real income. Therefore,

$$\widehat{W}_j = \begin{cases} \widehat{Y}_j / \widehat{P}_j^E, & j \in E, \\ \widehat{Y}_j / \widehat{P}_j^W, & j \in W. \end{cases} \tag{E.37}$$

E.10 Special case

The model nests the standard one-level CES gravity structure as a special case. If $\rho_E = \theta_E \equiv \theta$, the nested Eastern demand system collapses to a single CES aggregator over the domestic good and all foreign origins. In that case, the effective Eastern CES weights are β_j for the domestic good and $(1-\beta_j)\alpha_{ij}^E$ for foreign origin $i \neq j$. If, in addition, $\theta = \theta_W$, then Eastern and Western destinations are governed by the same one-level CES demand system, and the model reduces to the standard CES gravity specification.

E.11 Calibration of elasticities in the Western and Eastern Bloc

A central ingredient of the model is the elasticity structure governing substitution and trade responsiveness. For Western economies and third countries, we adopt a trade elasticity $\theta_W = 5$. This value is taken from Head and Mayer (2014) and is consistent with the standard estimates in the quantitative trade literature, which typically lie in the range of 4–8 (see, e.g., Anderson and van Wincoop, 2003; Eaton and Kortum, 2002; Simonovska and Waugh, 2014). In contrast, we assume substantially lower elasticities for the Eastern bloc, setting the substitution elasticity to $\rho_E = 2$ and the trade elasticity to $\theta_E = 3$.

These lower elasticities are intended to capture the institutional features of trade within the Council for Mutual Economic Assistance, where allocation was largely determined through centralized planning and bilateral negotiations rather than decentralized market clearing. A well-established literature emphasizes that trade flows within planned economies were only weakly responsive to relative prices, reflecting both administrative constraints and the limited role of profit-maximizing behavior (Gregory and Stuart, 2001; Kornai, 1992).

First, the choice of $\rho_E = 2$ reflects the idea that, while some substitution across suppliers was technologically feasible, it was not exploited very much in practice. Production and procurement decisions in centrally planned economies were characterized by rigid input allocations and limited flexibility, leading to persistent trade patterns and weak adjustment margins (Kornai, 1980; Berliner, 1976). In this sense, $\rho_E = 2$ provides a case that rules out the degree of responsiveness to prices observed in market economies.

Second, we set the trade (Armington) elasticity to $\theta_E = 3$, substantially below typical estimates for market economies. This choice reflects several features of East bloc trade. Transactions were frequently governed by bilateral agreements with quantities negotiated in advance, often in physical units, and subject to political considerations rather than relative prices (Marrese and Vanous, 1983; Hanson, 2003). Moreover, administered prices—often based on moving averages of world prices or entirely detached from contemporaneous market conditions—further reduced

the informational content of prices and dampened quantity responses. Together, these features imply that trade flows were significantly less sensitive to trade costs than in market-based systems.

The chosen value $\theta_E = 2$ should therefore be interpreted as a conservative representation of low, but non-zero, responsiveness. It allows for some reallocation in response to changes in trade costs while remaining well below Western benchmarks.

Overall, the combination $(\rho_E, \theta_E) = (2, 3)$ captures the key qualitative distinction between market and planned trade systems: while substitution and reallocation are not entirely absent in the Eastern bloc, they are markedly attenuated due to institutional rigidities, soft budget constraints, and the limited role of market-clearing prices.

We conduct robustness checks in which we select higher values for these two elasticities. In particular, we also report results for the limiting case in which the elasticities in the East coincide with the elasticity in the West, i.e., $(\rho_E, \theta_E, \theta_W) = (5, 5, 5)$.

E.12 Minimum distance inversion of the cross-bloc trade-cost component

This subsection describes a low-dimensional inversion procedure that maps observed changes in bilateral trade flows into a common cross-bloc trade-cost change. The procedure is used to calibrate the cross-bloc component in (E.36).

We begin with observed bilateral expenditure matrices for two periods, denoted by X^0 and X^1, defined over the core East–West country sample. Let $\widehat{\tau}_{ij}$ denote the trade-cost change from period 0 to period 1. We impose the same one-parameter structure as in (E.36):

$$\widehat{\tau}_{ij}(\widehat{b}) = \begin{cases} \widehat{b} & \text{if } i \in E,\ j \in W \text{ or } i \in W,\ j \in E, \\ 1 & \text{otherwise,} \end{cases}$$

where E and W denote the East and West blocs. Thus, the only friction allowed to change is the common cross-bloc iceberg cost.

For a given elasticity triple $(\rho_E, \theta_E, \theta_W)$, the model is solved in exact hats using the period-0 trade matrix as the benchmark equilibrium. Western destinations are modeled with a one-level CES demand system with elasticity θ_W. Eastern destinations are modeled with the nested CES structure in (E.2)–(E.7), in which θ_E governs substitution across foreign suppliers and ρ_E governs substitution between the domestic good and the foreign bundle. For any candidate value of $\widehat{b}$, the model delivers bilateral trade hats $\widehat{X}_{ij}^{model}(\widehat{b})$ and the associated equilibrium price and welfare changes.

To compare the model to the data, we do not target raw bilateral trade-flow changes. Instead, we use the symmetric Head-Ries transformation, which benchmarks bilateral trade against domestic trade in both directions. This transformation is natural for taking the model to the data because trade costs recovered from the empirical exercise have a symmetric structure. For an unordered country pair $\{i, j\}$ in period t, define

$$y_{ij,t} \equiv \frac{1}{2} \left[\log X_{ii,t} + \log X_{jj,t} - \log X_{ij,t} - \log X_{ji,t} \right].$$

The observed change in this measure is

$$\Delta y_{ij}^{data} = y_{ij,1} - y_{ij,0}.$$

This transformation differences out much of the country-size and multilateral-resistance variation that would otherwise contaminate direct comparisons of bilateral trade flows.

The model analogue is constructed from the simulated trade hats. Since the exact-hat system is normalized relative to the benchmark period,

$$\Delta y_{ij}^{model}(\widehat{b}) = \frac{1}{2} \left[\log \widehat{X}_{ii}^{model}(\widehat{b}) + \log \widehat{X}_{jj}^{model}(\widehat{b}) - \log \widehat{X}_{ij}^{model}(\widehat{b}) - \log \widehat{X}_{ji}^{model}(\widehat{b}) \right].$$

This gives the model-implied change in symmetric bilateral trade frictions for pair $\{i, j\}$.

The common cross-bloc barrier $\widehat{b}$ is then estimated by minimum distance:

$$\widehat{b} = \arg\min_{\widehat{b}} \sum_{\{i,j\} \in CB} \omega_{ij} \left(\Delta y_{ij}^{data} - \Delta y_{ij}^{model}(\widehat{b}) \right)^2,$$

where CB denotes the set of unordered cross-bloc country pairs and ω_{ij} are baseline trade-share weights. In the implementation, unsupported pairs are excluded, so the criterion is evaluated only when bilateral and domestic trade flows are observed in both periods.

This inversion procedure is deliberately restrictive. It does not attempt to recover a full matrix of bilateral trade-cost changes. Instead, it asks whether the observed evolution of cross-bloc trade frictions can be rationalized by a single common cross-bloc iceberg wedge within the exact-hat model. The resulting estimate $\widehat{b}$ can therefore be interpreted as the common cross-bloc trade-cost change that best matches the symmetric bilateral trade-cost signal in the data for a given elasticity configuration.

The Head-Ries transformation is exact in standard one-level Armington settings under symmetric bilateral trade costs. In the present framework, however, Eastern destinations are governed by a nested demand system. For that reason, the transformed bilateral measure is not treated as an exact inversion formula for the model. Instead, it is used as a minimum-distance target that

summarizes the observed evolution of cross-bloc trade frictions in a way that is closely related to the trade-cost object of interest.

E.13 Model fit and parameter sensitivity analysis

Tables E.2 and E.3 show that the main quantitative conclusions from the model simulations are robust to alternative elasticity choices. On the calibration side, the estimated cross-bloc trade-cost changes display a clear and economically coherent pattern across all specifications. For the transition from the early peak to the post-Détente trough, the implied cross-bloc barrier always declines substantially, with $\hat{\tau}^{CB} < 1$ in every case. For the transition from the post-Détente trough to the late peak, the estimated barrier always rises, with $\hat{\tau}^{CB} > 1$ throughout. The magnitude of the implied trade-cost change varies systematically with the elasticity assumptions in the expected direction: lower elasticities on the Eastern side require larger movements in $\hat{\tau}^{CB}$ to rationalize a given change in trade patterns, while higher elasticities imply more muted trade-cost changes. At the same time, the fit of the inversion remains fairly stable across specifications, with only modest variation in the RMSE values.

The welfare results are similarly robust. In every elasticity configuration, the move from the early peak to the post-Détente trough increases welfare in both blocs, while the reversal from the trough to the late peak reduces welfare in both blocs. More importantly, the asymmetry highlighted in the main text is preserved throughout: Eastern countries always experience larger gains from geopolitical relaxation and larger losses from renewed fragmentation than Western countries. The exact magnitudes vary with the elasticity assumptions, but the ordering of the effects is remarkably stable. This is especially informative because the comparison includes both the baseline case, which allows for reduced substitution on the Eastern side, and the reference case $(5, 5, 5)$, which removes that asymmetry. The fact that the same qualitative conclusions emerge in both environments suggests that the main findings are not driven by a narrow parametric choice, but instead reflect a robust feature of the historical evolution of cross-bloc trade frictions.

Table E.2: Cross-bloc trade-cost changes and fit across elasticity specifications

Transition	$(\rho_E, \theta_E, \theta_W)$	$\hat{\tau}^{CB}$	**RMSE**
Early peak → Trough	$(2, 3, 5)$	0.669	0.941
	$(1, 3, 5)$	0.634	0.941
	$(3, 3, 5)$	0.691	0.944
	$(2, 4, 5)$	0.722	0.943
	$(2, 2, 5)$	0.563	0.939
	$(2, 5, 5)$	0.755	0.945
	$(5, 5, 5)$	0.781	0.949
Trough → Late peak	$(2, 3, 5)$	1.187	0.407
	$(1, 3, 5)$	1.198	0.423
	$(3, 3, 5)$	1.175	0.401
	$(2, 4, 5)$	1.146	0.408
	$(2, 2, 5)$	1.287	0.403
	$(2, 5, 5)$	1.124	0.410
	$(5, 5, 5)$	1.114	0.400

Notes: The table reports the estimated cross-bloc trade-cost change, $\hat{\tau}^{CB}$, and the corresponding weighted root mean squared error (RMSE) from the Head-Ries minimum-distance objective for two benchmark transitions: from the early peak of fragmentation (1951–1953) to the post-Détente trough (1980–1982), and from the post-Détente trough (1980–1982) to the late peak (1986–1988). The baseline specification is $(\rho_E, \theta_E, \theta_W) = (2, 3, 5)$. The remaining rows report robustness cases and the reference case $(5, 5, 5)$.

Table E.3: Average welfare effects across elasticity specifications

Transition	$(\rho_E, \theta_E, \theta_W)$	**West mean**	**East mean**
Early peak → Trough	$(2, 3, 5)$	0.71	2.96
	$(1, 3, 5)$	0.67	3.20
	$(3, 3, 5)$	0.73	2.78
	$(2, 4, 5)$	0.66	2.12
	$(2, 2, 5)$	0.83	4.84
	$(2, 5, 5)$	0.64	1.65
	$(5, 5, 5)$	0.68	1.50
Trough → Late peak	$(2, 3, 5)$	-0.15	-0.60
	$(1, 3, 5)$	-0.14	-0.52
	$(3, 3, 5)$	-0.15	-0.60
	$(2, 4, 5)$	-0.14	-0.39
	$(2, 2, 5)$	-0.15	-1.11
	$(2, 5, 5)$	-0.14	-0.29
	$(5, 5, 5)$	-0.14	-0.31

Notes: The table reports average welfare changes, in percent, for Western and Eastern countries under the same two benchmark transitions used in the main text: from the early peak of fragmentation (1951–1953) to the post-Détente trough (1980–1982), and from the post-Détente trough (1980–1982) to the late peak (1986–1988). Welfare is evaluated on the common post-Détente trough trade structure. Reported values are simple arithmetic averages across countries within each bloc. The baseline specification is $(\rho_E, \theta_E, \theta_W) = (2, 3, 5)$; the remaining rows report robustness cases and the reference case $(5, 5, 5)$.